

SCHMIDT CIRCLES OF THE GAUSSIAN INTEGERS: CLASSIFICATION, CLASS NUMBERS, AND THE PHASE INVARIANT

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ABSTRACT. The Gaussian Schmidt arrangement is the orbit of the extended real line under the Bianchi group, $\mathcal{S} = \mathrm{PSL}_2(\mathbb{Z}[i]) \cdot \widehat{\mathbb{R}}$: an arrangement of circles first drawn by Asmus Schmidt and studied by Stange. We develop its arithmetic in three layers. First, starting from the elementary congruence classification of its circles, we stratify $\mathcal{S}$ by an integer level α — simultaneously the hyperbolic radius invariant $\coth \rho$, a trace on the Cartan embedding $X \mapsto X\bar{X}^{-1}$, and the inversive product with $\widehat{\mathbb{R}}$ — and count the arrangement in its three background geometries: per unit area the circles of curvature $2n$ number $N_e(n) = n \prod_{p|n} (1 - \chi_{-4}(p)/p) = 2h(-4n^2)$, with mean value governed by Catalan’s constant; the circles of level n in an ideal triangle of the hyperbolic plane have weighted count $3H(n^2 - 1)$; and on the Riemann sphere the circles of spherical radius $\arccot \ell$ number $\frac{1}{3}r_3(\ell^2 + 1) = 4H(4(\ell^2 + 1))$. Second, we identify the involution $\sigma(X) = \bar{X}^{-1}$ as the unitary involution adjoint to the Hermitian form of $\widehat{\mathbb{R}}$, its Cartan image as the inversion in the circle $X(\widehat{\mathbb{R}})$, and prove the class formula: on the primitive form classes of discriminant $1 - n^2$ attached to level- n circles, σ acts by $[f] \mapsto [\mathfrak{r}_n][f]^{-1}$ for a canonical ambiguous class $\mathfrak{r}_n$ — formally the Atkin–Lehner involution on Heegner points — and Gauss composition itself becomes a geometric calculus of mirrors and magnifications of circles. Third, we construct the sixth coordinate on the double-coset space $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{C}) / \mathrm{SL}_2(\mathbb{Z})$ — a phase — and show that on Schmidt circles it quantizes: the normalized phase u_f equals the derivative $\Phi_y / \Phi_x(\beta_1, \beta_2)$ of the modular correspondence $X_0(\frac{n-1}{2})$ at a Heegner pair whose discriminant is coupled to the level, obeys a generalized dihedral Galois law at first power, and generates one integer polynomial per level, irreducible at every computed level, with denominators supported at Gross–Zagier primes; on ambiguous divisor classes its sign obeys a proved archimedean law and is provably not a genus character. A parallel Euclidean theory (translations in place of one modular factor), normalized by the lemniscatic period, produces monic integer level polynomials irreducible for *every* level, and an exact Δ -mass law. Certified high-precision computations show the phase is not a rational linear combination of the Duke–Imamoğlu–Tóth cycle-integral invariants of its coupled real quadratic discriminant: the phase appears to be a genuinely new invariant. Every finite claim in the paper is machine-verified; the scripts and precision policies are catalogued in an appendix.

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1. INTRODUCTION

Let $\Gamma_B = \mathrm{PSL}_2(\mathbb{Z}[i])$ act on $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ by Möbius transformations. The *Gaussian Schmidt arrangement* is the orbit of the extended real line,

$$(1.1) \quad \mathcal{S} = \Gamma_B \cdot \widehat{\mathbb{R}},$$

a countable union of circles and lines that first appeared, as the boundary data of a complex Farey subdivision, in Asmus Schmidt's theory of complex continued fractions [18], and whose systematic study was begun by Stange [20, 21]: for an arbitrary imaginary quadratic field the analogous orbit is an intricate circle arrangement whose tangencies, curvatures and connectivity encode the arithmetic of the field, and for $\mathbb{Q}(i)$ it coincides, up to scaling, with the Apollonian super-packing of Graham, Lagarias, Mallows, Wilks and Yan [5, 20]. Figure 3 shows a portion.

This paper develops the arithmetic of $\mathcal{S}$ in three layers, of increasing depth.

First layer: classification and three censuses. The circles of $\mathcal{S}$ admit an elementary Diophantine description (Theorem 2.1; in equivalent forms this is known [5, 20, 16]): a circle of radius $\frac{1}{2n}$ centered at $\zeta/2n$ belongs to $\mathcal{S}$ if and only if $\zeta = x + iy \in \mathbb{Z}[i]$ with x even, y odd and $x^2 + y^2 \equiv 1 \pmod{4n}$ — all conditions together amounting to the single Diophantine equation $u^2 + v(v+1) = nm$. Our starting point is a stratification of $\mathcal{S}$ that does not appear in the literature: the integer

$$\alpha = \frac{\text{height of center}}{\text{radius}}$$

of a circle in the upper half-plane, which is simultaneously coth of its hyperbolic radius, the trace invariant $-\frac{1}{2} \mathrm{tr}(X\bar{X}^{-1})$ of the Cartan embedding, and the inversive product of the circle with $\widehat{\mathbb{R}}$ (Proposition 2.4). The level- n circles are in explicit bijection with the positive definite binary quadratic forms of discriminant $1 - n^2$, their hyperbolic centers are the CM points of that discriminant, and their hyperbolic radius $\mathrm{arccoth} n$ depends on the level alone (Proposition 2.5).

Counting the arrangement then produces class numbers in each of the three classical geometries carried by $\widehat{\mathbb{C}}$ (Theorem 3.8): per unit area of $\mathbb{C}$ the circles of curvature $2n$ number

$$N_e(n) = n \prod_{p|n} \left(1 - \frac{\chi_{-4}(p)}{p}\right) = 2h(-4n^2), \quad \sum_{n \leq X} N_e(n) = \frac{X^2}{2G} + O(X \log X),$$

with $G = L(2, \chi_{-4})$ *Catalan's constant* (Propositions 3.1–3.2; the class-number form of the count is, for $\mathbb{Q}(i)$, a theorem of Stange [21], which we

reprove through an explicit lattice dictionary used heavily later); the circles of level n whose hyperbolic center lies in the ideal triangle $(0, 1, \infty)$, counted with weight $\frac{1}{2}$ on the edges, number $3H(n^2 - 1)$, where H is the Hurwitz class number (Theorem 3.3); and on the Riemann sphere the spherical radii quantize to the integer-cotangent angles $\cot \theta = \ell$, with exactly

$$N_{\text{sph}}(\ell) = \frac{1}{3} r_3(\ell^2 + 1) = 4H(4(\ell^2 + 1))$$

circles of spherical radius $\text{arccot } \ell$ (Theorem 3.6). One arrangement, three geometries, three class-number censuses; the hyperbolic and spherical counts read the Hurwitz–Cohen–Eisenstein weight-3/2 coefficients along the two quadratic progressions $d = n^2 - 1$ and $d = 4\ell^2 + 4$, complementary to the Euclidean $d = 4n^2$.

Second layer: the involution and the class formula. The map $\sigma(X) = \bar{X}^{-1}$ is the adjoint anti-involution of the Hermitian form defining $\widehat{\mathbb{R}}$ — a unitary involution, of the second kind, on $M_2(\mathbb{Q}(i))$ — and its Cartan image $X\bar{X}^{-1}$ is, exactly, the anti-holomorphic inversion in the circle $X(\widehat{\mathbb{R}})$ (Theorem 4.2); the trace identity $\text{tr}(X\bar{X}^{-1}) = -2\alpha$ recovers the level and attaches to every level- n circle a closed geodesic of length $2 \text{arccosh } n$ in the Bianchi orbifold. On the level- n form classes, σ induces an involution $\hat{\sigma}$, and our first main arithmetic result computes it (Theorem 4.5): on primitive classes

$$\hat{\sigma}[f] = [\mathfrak{r}_n][f]^{-1}, \quad \mathfrak{r}_n = \left[\left(\frac{n-1}{2}, 0, \frac{n+1}{2} \right) \right],$$

a Galois conjugation of the class group twisted by a canonical ambiguous class — formally identical to the Atkin–Lehner action $[\mathfrak{a}] \mapsto [\mathfrak{n}\bar{\mathfrak{a}}]$ on Heegner points. The proof runs through an explicit closed-form section (Lemma 4.6): a formula producing, for every prescribed level- n circle, a matrix $X \in \text{SL}_2(\mathbb{Z}[i])$ realizing it — a tool we expect to be of independent use. Composition of forms itself becomes visible in the arrangement: the inverse of a circle class is its mirror image, Gauss composition is a Chinese-remainder gluing characterized by integer magnifications, and 2-torsion classes are exactly the circles with a mirror symmetry (Section 5).

Third layer: the phase. The pair of circles $(X(\widehat{\mathbb{R}}), \sigma(X)(\widehat{\mathbb{R}}))$ has five real invariants under $X \mapsto \gamma X \gamma'$, $\gamma, \gamma' \in \text{SL}_2(\mathbb{Z})$: the level α and the two singular moduli β_1, β_2 at the hyperbolic centers. The sixth coordinate on the double-coset space $\text{SL}_2(\mathbb{Z}) \backslash \text{SL}_2(\mathbb{C}) / \text{SL}_2(\mathbb{Z})$ is a *phase* $\Theta(X)$, built from a weight-2 kernel and the derivative of X at a distinguished intertwined CM point (Section 6). On the Bianchi group this continuous invariant quantizes to algebraic numbers $u_f = \varepsilon \Theta_f$, $\varepsilon = n + \sqrt{n^2 - 1}$, indexed by the form classes of the level, and obeys the exact laws $u_{f-1} = \bar{u}_f$ and $u_f u_{\mathfrak{r}_n f} = 1$. The pair (β_1, β_2) of a class and its twist satisfies the *simultaneous modular equations* $\Phi_{(n-1)/2}(\beta_2, \bar{\beta}_1) = 0 = \Phi_{(n+1)/2}(\beta_2, \bar{\beta}_1)$: each $\hat{\sigma}$ -pair of Schmidt circles is a point on the fiber product $X_0(\frac{n-1}{2}) \times_{X(1)} X_0(\frac{n+1}{2})$ whose discriminant

is coupled to its level — and the phase is precisely the derivative of this correspondence at its Heegner pair:

$$u_f = \frac{\Phi_y(\beta_1, \beta_2)}{\Phi_x(\beta_1, \beta_2)}, \quad \Phi = \Phi_{(n-1)/2}$$

(Theorem 6.19). Because the right-hand side is a rational expression with rational coefficients in two singular moduli, the Galois action on u_f is completely transparent: u_f lies in the ring class field of discriminant $1 - n^2$ and transforms under the full generalized dihedral law *at first power*, with no root-of-unity cocycle (Theorem 6.19). Consequences: the h phases of a level are the roots of a single integer polynomial Q_n of degree $h(1 - n^2)$, irreducible at every computed level $n \leq 17$ (Theorems 6.21–6.22); the denominators of u_f are supported at the Gross–Zagier primes of the discriminant pairs $(1 - n^2, -3)$ and $(1 - n^2, -4)$ — the primes where the level’s CM points collide with the two elliptic points of the modular orbifold (Section 6.7); and on ambiguous divisor classes the *sign* of u_f obeys a proved archimedean law — a gcd inequality reading off on which side of the E_6 -zero the twisted CM point falls — and is provably *not* a homomorphism on $\text{Cl}[2]$ (Theorem 6.28, Proposition 6.27).

A parallel *Euclidean* phase theory, in which the left $\text{SL}_2(\mathbb{Z})$ -factor is replaced by the translations $\mathbb{Z}[i]$, is developed in Section 7: there the level’s CM data is the ring class field of the order $\mathbb{Z} + n\mathbb{Z}[i]$ (conductor equal to the level, field fixed), the phase is normalized by the lemniscatic period $\Omega = \varpi^2/\pi$, and the theory is complete in a way the hyperbolic one is not yet: the level polynomials are *monic* with integer coefficients, and are irreducible *for every level* (Theorem 7.16), by a cocycle-free Shimura translation law at square level and an archimedean dominance argument; moreover the total Δ -mass of a level obeys an exact product law over the non-split primes (Theorem 7.18).

What is new, and what is not. The classification of Section 2 is essentially known: curvature integrality and the Hermitian framework are Stange’s [21], and the congruence description is equivalent to the strongly-integral super-packing description of [5] combined with the identification of $\mathcal{S}$ with the super-packing [20]; a congruence form is stated in [16]. We include a short self-contained proof to fix normalizations and because the descent it sets up is used throughout. Likewise the Euclidean census in its class-number form specializes a theorem of Stange [21]. Everything else, to the best of our knowledge — the level stratification and the hyperbolic and spherical censuses, the identification of σ and the class formula, the circle calculus of composition, the six-coordinate moduli system, the phase and all its laws, and the Euclidean phase theory — is new. Three literature probes are recorded in the repository accompanying this paper; beyond the Schmidt-arrangement literature proper [18, 20, 21, 16, 14], the objects closest in spirit are the cycle-integral invariants of real quadratic discriminants

of Duke, Imamoğlu and Tóth [3] attached, through the Katok–Sarnak correspondence [10], to the same coupled discriminant pairs $(1 - n^2, 4(n^2 - 1))$ that index our phase. This proximity can be tested: at levels $3 \leq n \leq 17$, certified integer-relation searches at 150-digit precision (with conservative safety margins) find *no* rational linear relation between $\log |u_f|$, $\arg u_f$ or the pair-sums $u_f + u_f^{-1}$ and the discriminant- $4(n^2 - 1)$ cycle integrals of j , the Kronecker-limit integrals, $\log \varepsilon_n$, π , π^2 and 1 (Section 6.9). The phase is, as far as we can measure, genuinely new real-quadratic-adjacent data.

Status conventions. Every statement below is one of two kinds, and the distinction is maintained explicitly. A **Theorem**, **Proposition**, **Lemma** or **Corollary** without qualification is proved. A statement labelled (*computational*) is a finite claim verified by exact arithmetic, or a numerical claim certified at a stated precision with stated safety margins, by a named script; the range of verification is part of the statement. Appendix A lists every script cited, what it verifies, at what precision, and its runtime; all scripts were re-run for this paper. We never state as a theorem anything whose proof is incomplete, and we flag open ends where they occur (they are collected in Section 8).

Plan of the paper. Section 2 fixes the Hermitian encoding, proves the classification, and introduces the level. Section 3 proves the three counting theorems. Section 4 identifies σ and proves the class formula. Section 5 develops composition in circle language. Section 6 constructs the phase, proves its laws, the first-power descent and the Galois theory, the level polynomials, the Gross–Zagier support, and the sign law, and records the Duke–Imamoğlu–Tóth comparison. Section 7 develops the Euclidean phase. Section 8 collects the open problems. Appendix A documents the machine verification.

Acknowledgements. The results of this paper were obtained in a sequence of computer-assisted research sessions; the complete research notes, verification scripts and figures are available in the accompanying repository.

2. CLASSIFICATION AND THE LEVEL

2.1. Circles as Hermitian matrices. A circle or line in $\mathbb{C}$ is the zero set of $Az\bar{z} + \bar{B}z + B\bar{z} + C = 0$ with $A, C \in \mathbb{R}$, $B \in \mathbb{C}$ and $|B|^2 - AC > 0$, encoded by the Hermitian matrix

$$M = \begin{pmatrix} A & B \\ \bar{B} & C \end{pmatrix}, \quad z \in \text{circle} \iff (\bar{z}, 1) M \begin{pmatrix} z \\ 1 \end{pmatrix} = 0.$$

For $A \neq 0$ this is the circle of center $z_0 = -B/A$ and radius $r = \sqrt{-\det M}/|A|$; for $A = 0$ it is a line. The matrix M determines the circle and is determined by it up to a real scalar; oriented circles correspond to M up to a *positive*

scalar, and $M \mapsto -M$ reverses the orientation. The extended real line is $\text{Im } z = 0$, i.e.

$$(2.1) \quad M_0 = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad \det M_0 = -1.$$

If $w = gz$ then $(z, 1)^\top \propto g^{-1}(w, 1)^\top$, so the image circle $g \cdot \mathcal{C}$ has matrix

$$(2.2) \quad M_{g\mathcal{C}} = (g^{-1})^\dagger M_{\mathcal{C}} g^{-1},$$

and for $g \in \text{SL}_2$ the determinant is preserved. Hence $\mathcal{S}$ is in bijection with the $\text{PSL}_2(\mathbb{Z}[i])$ -orbit of M_0 inside integral Hermitian matrices of determinant -1 , up to sign for unoriented circles; every circle of $\mathcal{S}$ has radius $1/|A|$. Since $z \mapsto -z$ (the class of $\text{diag}(i, -i)$) lies in $\text{PSL}_2(\mathbb{Z}[i])$ and reverses the orientation of $\widehat{\mathbb{R}}$, each circle of $\mathcal{S}$ occurs with both orientations, and we may normalize $A \geq 0$.

For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}[i])$ a two-line computation from (2.2) gives

$$(2.3) \quad (g^{-1})^\dagger M_0 g^{-1} = \begin{pmatrix} \frac{2 \text{Im}(cd)}{i(ad - b\bar{c})} & i(ad - b\bar{c}) \\ i(ad - b\bar{c}) & 2 \text{Im}(ab) \end{pmatrix}.$$

So the image $g \cdot \widehat{\mathbb{R}}$ has *even integer* curvature $A = 2 \text{Im}(cd)$ and co-curvature $C = 2 \text{Im}(ab)$, and $B = i(ad - b\bar{c}) \in \mathbb{Z}[i]$. Writing the curvature as $2n$ ($n \geq 1$; $n = 0$ gives the lines) we define the *curvature-center*

$$\zeta := 2n z_0 = -B \in \mathbb{Z}[i], \quad \zeta = x + iy,$$

so the circle has radius $\frac{1}{2n}$ and center $\zeta/2n$.

2.2. The classification.

Theorem 2.1 (classification). *A circle of radius r centered at z_0 belongs to $\mathcal{S} = \text{PSL}_2(\mathbb{Z}[i]) \cdot \widehat{\mathbb{R}}$ if and only if*

$$r = \frac{1}{2n} \quad (n \in \mathbb{Z}_{\geq 1}), \quad \zeta := 2n z_0 = x + iy \in \mathbb{Z}[i],$$

with

$$x \equiv 0, \quad y \equiv 1 \pmod{2}, \quad x^2 + y^2 \equiv 1 \pmod{4n}.$$

The lines of $\mathcal{S}$ are exactly $\text{Im } z = k$, $k \in \mathbb{Z}$.

In equivalent forms this statement is known: it amounts to the description of the strongly integral Apollonian super-packing of [5, §6] combined with the identification of $\mathcal{S}$ with the super-packing [20, Thm. 1.3]; a congruence form appears in [16], and curvature integrality and the framework are due to Stange [21]. We include the short proof because the normalization and the descent it sets up are used throughout the paper.

Proof. Necessity. (i) $\det M = -1$ reads $4nm - |\zeta|^2 = -1$ where $C = 2m$, i.e. $x^2 + y^2 = 1 + 4nm$; the auxiliary m is not free data: $2m$ is the co-curvature, determined by (n, ζ) . (ii) Reduce modulo 2 in $\mathbb{Z}[i]$. Since $\bar{\alpha} \equiv \alpha \pmod{2}$ for every $\alpha \in \mathbb{Z}[i]$, we get $a\bar{d} - b\bar{c} \equiv ad - bc = 1 \pmod{2}$, so $B \equiv i \pmod{2}$;

this is invariant under $B \mapsto -B$, and $\zeta = -B$ satisfies $\zeta \equiv i \pmod{2}$, i.e. x even, y odd. (iii) Primitivity of $(2n, \zeta, 2m)$ is automatic: a common divisor divides $|\zeta|^2 - 4nm = 1$. For the lines: if $A = 2\operatorname{Im}(c\bar{d}) = 0$ with $\gcd(c, d) = 1$, then $d/c \in \mathbb{Q}$ forces $c = qu$, $d = pu$ with p, q coprime rational integers and u a unit; then $ad - bc = u(ap - bq) = 1$ gives $B = i\bar{u}(ap - bq) = i\bar{u}^2 = \pm i$, so the line is $\operatorname{Im} z = k$ with $k \in \mathbb{Z}$. Vertical lines do not occur.

Sufficiency. Let $M = \begin{pmatrix} 2n & -\zeta \\ -\zeta & 2m \end{pmatrix}$ be integral Hermitian with $\det M = -1$, $n \geq 1$, $\zeta \equiv i \pmod{2}$. Two moves are available inside $\operatorname{PSL}_2(\mathbb{Z}[i])$, both preserving the constraint set: *translate* ($z \mapsto z + \lambda$, $\lambda \in \mathbb{Z}[i]$), which replaces $\zeta \mapsto \zeta + 2n\lambda$ and preserves the parity class; and *invert* ($z \mapsto -1/\bar{z}$), which swaps $A \leftrightarrow C$ and sends $B \mapsto -\bar{B}$, preserving $B \equiv i \pmod{2}$. Translate so that $\operatorname{Re} \zeta, \operatorname{Im} \zeta \in (-n, n]$; then $|\zeta|^2 \leq 2n^2$ and

$$|2m| = \frac{||\zeta|^2 - 1|}{2n} \leq \frac{2n^2 - 1}{2n} < n < 2n.$$

If $m \neq 0$, invert (and negate M if needed to restore positive curvature): the curvature strictly drops from $2n$ to $|2m| < n$. A strictly decreasing sequence of positive even integers terminates, so we reach $m = 0$; then $|\zeta|^2 = 1$ and $\zeta \equiv i \pmod{2}$ force $\zeta = \pm i$: the circle of radius $\frac{1}{2n}$ tangent to $\widehat{\mathbb{R}}$ at 0. One more inversion sends it to the line $\operatorname{Im} z = \mp n'$, a translate of $\widehat{\mathbb{R}}$. Reversing the walk exhibits M in the orbit. $\square$

Corollary 2.2 (single equation). *Substituting $x = 2u$, $y = 2v+1$, the circles of $\mathcal{S}$ are exactly*

$$\left\{ \left| z - \frac{2u+(2v+1)i}{2n} \right| = \frac{1}{2n} : u, v, n, m \in \mathbb{Z}, n \geq 1, u^2 + v(v+1) = nm \right\},$$

one circle per solution; $m = (u^2 + v^2 + v)/n$ is determined, and $2m$ is the co-curvature.

Remark 2.3 (the PGL_2 variant). $\operatorname{PGL}_2(\mathbb{Z}[i])/\operatorname{PSL}_2(\mathbb{Z}[i]) \cong \mathbb{Z}/2$, generated by $z \mapsto iz$. Repeating the parity computation with $\det g = \pm i$ gives $\zeta \equiv \pm i \pmod{2}$ (x odd, y even), whence

$$\operatorname{PGL}_2(\mathbb{Z}[i]) \cdot \widehat{\mathbb{R}} = \mathcal{S} \sqcup i\mathcal{S},$$

a *disjoint* union: the two families share no circle and no line ($i\mathcal{S}$ contains the vertical lines). The PGL_2 -arrangement is the one with fourfold symmetry; $\mathcal{S}$ itself is invariant only under $z \mapsto z + \mathbb{Z}[i]$, $z \mapsto -z$, $z \mapsto \bar{z}$, and $\operatorname{PSL}_2(\mathbb{Z}[i])$. In the PGL_2 -arrangement the classification reads: $x+y$ odd and $x^2 + y^2 \equiv 1 \pmod{4n}$, and all counting functions double.

Machine verification: the script `verify_classification.py` enumerates the orbit of M_0 by breadth-first search over bounded Hermitian matrices and confirms exact set equality between the orbit's residue classes ($\zeta \pmod{2n}$) and the congruence classes of Theorem 2.1 for all $n \leq 15$, stable across cutoffs, along with the parity invariant and the horizontal lines (Appendix A).

2.3. The level. From now on we single out the circles of the arrangement $\mathcal{S} \sqcup i\mathcal{S}$ contained in the open upper half-plane $\mathbb{H}$.

Proposition 2.4 (the level and its three faces). *For a circle $\omega \subset \mathbb{H}$ with Euclidean center $x_0 + iy_0$ and radius r , set $\alpha(\omega) = y_0/r$. Then:*

- (1) α is invariant under $\mathrm{PSL}_2(\mathbb{R})$; in fact $\alpha(\omega) = \coth \rho_{\mathrm{hyp}}(\omega)$, where ρ_{hyp} is the hyperbolic radius of ω as a circle of the hyperbolic plane $\mathbb{H}$.
- (2) $\alpha(\omega)$ is the inversive product of ω with $\widehat{\mathbb{R}}$: for circles with Hermitian matrices M, N normalized to $\det = -1$, the inversive product is $\langle M, N \rangle = \frac{1}{2} \mathrm{tr}(MN^{\mathrm{adj}})$ with N^{adj} the adjugate, and $\langle M_\omega, M_0 \rangle = \alpha(\omega)$.
- (3) If $\omega = X(\widehat{\mathbb{R}})$ with $X \in \mathrm{SL}_2(\mathbb{C})$, then $\mathrm{tr}(X\bar{X}^{-1}) = -2\alpha_\pm(\omega)$, the sign recording the orientation (Theorem 4.2 below).

On Schmidt circles, α is an integer $n \geq 2$: for a circle of curvature $2q$ and curvature-center $x + iy$ one has $\alpha = y$. Odd levels belong to $\mathcal{S}$ and even levels to $i\mathcal{S}$; the value $\alpha = 1$ corresponds to circles tangent to $\widehat{\mathbb{R}}$ (the Ford circles), which we exclude.

Proof. (1) The hyperbolic circle of hyperbolic center $a + bi$ and hyperbolic radius ρ is the Euclidean circle of center $a + ib \cosh \rho$ and radius $b \sinh \rho$; hence $\alpha = \coth \rho$, manifestly a hyperbolic isometry invariant. (2) With M_ω normalized as in §2.1, $M_\omega = \begin{pmatrix} 1/r & -z_0/r \\ -\bar{z}_0/r & (|z_0|^2 - r^2)/r \end{pmatrix}$, and for $M = (A_1, B_1, C_1)$, $N = (A_2, B_2, C_2)$ one computes $\frac{1}{2} \mathrm{tr}(MN^{\mathrm{adj}}) = \frac{1}{2}(A_1C_2 + C_1A_2) - \mathrm{Re}(B_1\bar{B}_2)$, which for two proper circles is the classical inversive product $\frac{|z_1 - z_2|^2 - r_1^2 - r_2^2}{2r_1r_2}$ and for $N = M_0$ evaluates to y_0/r . (3) is proved in Theorem 4.2. Finally for a Schmidt circle, $\alpha = \frac{y/2q}{1/2q} = y \in \mathbb{Z}$, and the parity of x (even for $\mathcal{S}$, odd for $i\mathcal{S}$) together with $x^2 + y^2 \equiv 1 \pmod{4}$ forces the parity of y , hence the parity split of the levels. $\square$

Proposition 2.5 (level- n circles are quadratic forms). *Fix $n \geq 2$ and write $N = n^2 - 1$, $D = 1 - n^2$. The circles with $\alpha = n$ are exactly*

$$\omega_{q,x} : \quad \text{radius } \frac{1}{2q}, \quad \text{center } \frac{x+ni}{2q}, \quad x^2 \equiv D \pmod{4q}, \quad q \geq 1,$$

and $\omega_{q,x} \mapsto f_\omega = (q, -x, m)$, $m = (x^2 + N)/4q$, is a bijection onto the positive definite integral binary quadratic forms of discriminant D (imprimitive forms included), equivariant for the actions of $\mathrm{PSL}_2(\mathbb{Z})$ on circles and on forms. Moreover:

- (1) the hyperbolic center of $\omega_{q,x}$ is the root $z_{\mathrm{hyp}} = \frac{x+i\sqrt{N}}{2q} \in \mathbb{H}$ of f_ω — a CM point of discriminant D ;
- (2) the hyperbolic radius is $\mathrm{arcoth} n$, depending on the level alone.

Consequently: the circles of the PGL_2 -arrangement contained in $\mathbb{H}$ are precisely the hyperbolic circles of radius $\mathrm{arcoth} n$ centered at the CM points of discriminant $1 - n^2$, for $n = 2, 3, 4, \dots$

Proof. By Theorem 2.1 (and Remark 2.3 for even levels), the level- n circles are the stated $\omega_{q,x}$, and f_ω has discriminant $x^2 - 4qm = D$, positive definite since $q > 0$. The Möbius action of $\mathrm{PSL}_2(\mathbb{Z})$ on Hermitian matrices (2.2) restricts, on the real parts, to the standard $\mathrm{SL}_2(\mathbb{Z})$ -action on forms, giving equivariance. (1) The hyperbolic center is $a + bi$ with $a = \frac{x}{2q}$ and $b = \sqrt{(n/2q)^2 - (1/2q)^2} = \frac{\sqrt{N}}{2q}$, which is the root of $qz^2 - xz + m$ in $\mathbb{H}$. (2) $\coth \rho = n$. $\square$

3. COUNTING IN THREE GEOMETRIES

The sphere $\widehat{\mathbb{C}}$ carries the three classical two-dimensional geometries: the Euclidean plane $\mathbb{C}$, the hyperbolic plane $\mathbb{H}$, and the round sphere itself. The Schmidt arrangement has a natural census in each, and each census is a class-number count. We prove the three counts and then state the triptych as one theorem.

3.1. The Euclidean census and Catalan's constant. Since $\mathcal{S}$ is invariant under translation by $\mathbb{Z}[i]$, the natural density of curvature- $2n$ circles is the number per unit area:

$$N_e(n) := \#\{\mathcal{C} \in \mathcal{S} : \text{curv} = 2n, \text{ center} \in [0, 1) + [0, 1)i\}.$$

Centers in the unit square correspond to residues $\zeta \bmod 2n$, and Theorem 2.1 identifies

$$(3.1) \quad N_e(n) = \#\{(x, y) \in (\mathbb{Z}/2n)^2 : x \text{ even, } y \text{ odd, } x^2 + y^2 \equiv 1 \pmod{4n}\}.$$

Proposition 3.1 (exact count).

$$N_e(n) = n \prod_{\substack{p|n \\ p \text{ odd}}} \left(1 - \frac{\chi_{-4}(p)}{p}\right), \quad \chi_{-4}(p) = (-1)^{(p-1)/2}.$$

In particular N_e is multiplicative, with $N_e(2^a) = 2^a$ and $N_e(p^a) = p^{a-1}(p - \chi_{-4}(p))$ for odd p .

Proof. Let $r(q) = \#\{(x, y) \bmod q : x^2 + y^2 \equiv 1\}$. Lifting (3.1) to $(\mathbb{Z}/4n)^2$ multiplies the unconstrained count by 4, and any solution mod 4 has exactly one odd coordinate, the swap $(x, y) \mapsto (y, x)$ being a parity-reversing involution; so the parity constraint halves the count and $N_e(n) = \frac{1}{8}r(4n)$. The function r is multiplicative by CRT. For odd p the smooth conic $x^2 + y^2 = 1$ over $\mathbb{F}_p$ has $p - \chi_{-4}(p)$ points, and Hensel lifting gives $r(p^a) = p^{a-1}(p - \chi_{-4}(p))$. For $p = 2$ one checks $r(2^k) = 2^{k+1}$ for $k \geq 2$ (directly for $k = 2, 3$, then by the 2-adic lifting argument; verified for $k \leq 11$ by `verify_classification.py`). Assembling with $4n = 2^{e+2}n'$, n' odd, gives the closed form. $\square$

Proposition 3.2 (mean count).

$$\sum_{n \leq X} N_e(n) = \frac{X^2}{2G} + O(X \log X), \quad G = L(2, \chi_{-4}) = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \dots$$

(Catalan's constant). *Equivalently: per unit area, the number of circles of $\mathcal{S}$ of curvature at most κ is $\sim \kappa^2/(8G)$, and the number of radius $\geq \rho$ is $\sim 1/(8G\rho^2)$.*

Proof. Since $\chi_{-4}(2) = 0$, the Dirichlet series $\sum N_e(n)n^{-s}$ has Euler product $\zeta(s-1)/L(s, \chi_{-4})$, and expanding $1/L$ gives the Möbius identity $N_e(n) = \sum_{d|n} \mu(d) \chi_{-4}(d) n/d$ (both sides multiplicative; check on prime powers). Hence

$$\sum_{n \leq X} N_e(n) = \sum_{d \leq X} \mu(d) \chi_{-4}(d) \sum_{q \leq X/d} q = \frac{X^2}{2} \sum_{d=1}^{\infty} \frac{\mu(d) \chi_{-4}(d)}{d^2} + O(X \log X),$$

and $\sum_d \mu(d) \chi_{-4}(d) d^{-2} = L(2, \chi_{-4})^{-1}$. $\square$

Here Catalan's constant enters purely arithmetically, through the local solution densities of $x^2 + y^2 \equiv 1 \pmod{4n}$ — equivalently through $\zeta_{\mathbb{Q}(i)}(2) = \zeta(2)L(2, \chi_{-4})$. The same constant is the volume quantum of the Bianchi orbifold ($\text{vol}(\text{PSL}_2(\mathbb{Z}[i]) \backslash \mathbb{H}^3) = G/3$); the agreement is no coincidence, but no hyperbolic geometry is needed above. Two easy by-products: the circles of curvature $2n$ in the unit square have total length $\pi \prod_{p|n} (1 - \chi_{-4}(p)/p) \asymp \pi$ for every n ; and the total disk area per unit area up to curvature $2X$ is $\sim \frac{\pi}{4G} \log X$, quantifying logarithmically the well-known density of the arrangement.

The census has a second, structural form: $N_e(n) = 2h(-4n^2)$, the class number of the order $\mathbb{Z} + n\mathbb{Z}[i]$. For $K = \mathbb{Q}(i)$ this is a theorem of Stange [21] (circles of curvature $f\sqrt{-\Delta}$, up to translation and rotation by 180° , are in bijection with $\ker(\text{Pic}(\mathcal{O}_f) \rightarrow \text{Pic}(\mathcal{O}_K))$, which for class number one is all of $\text{Pic}(\mathcal{O}_f)$). We give an explicit lattice-theoretic proof in Section 7 (Theorem 7.6), where the resulting dictionary — conductor equal to the curvature — carries the Euclidean phase theory.

3.2. The hyperbolic census: $3H(n^2 - 1)$. Recall the *Hurwitz class number* $H(N)$ for $N \equiv 0, 3 \pmod{4}$: the number of $\text{SL}_2(\mathbb{Z})$ -classes of positive definite integral binary quadratic forms of discriminant $-N$, imprimitive forms included, each class weighted by $1/s$, where $s \in \{1, 2, 3\}$ is the order of its stabilizer in $\text{PSL}_2(\mathbb{Z})$ [1, 22]. Note $N = n^2 - 1 \equiv 0 \pmod{4}$ for n odd and $\equiv 3 \pmod{4}$ for n even, so $H(n^2 - 1)$ is always of the admissible type.

Let T be the closed ideal triangle with vertices $0, 1, \infty$: the region $0 \leq \text{Re } z \leq 1$ outside the open disk $|z - \frac{1}{2}| < \frac{1}{2}$. We locate a circle by its hyperbolic center; with $z_{\text{hyp}} = \frac{x+i\sqrt{N}}{2q}$ as in Proposition 2.5,

$$z_{\text{hyp}} \in T \iff 0 \leq x \leq 2q \text{ and } x(2q - x) \leq n^2 - 1,$$

with the center on an edge of T exactly when one of the constraints is an equality; since $x(2q - x) = N \iff x = 2m$, the three edge conditions are $x = 0$, $x = 2q$, $x = 2m$. For even n , x is odd and no edge condition can hold: every circle is interior. For odd n , edge circles are ubiquitous.

Theorem 3.3 (hyperbolic census). *For every integer $n \geq 2$, the number of Schmidt circles ω with $\alpha(\omega) = n$ and hyperbolic center in T , counted with weight $\frac{1}{2}$ when the center lies on an edge of T , equals $3H(n^2 - 1)$.*

Proof. By Proposition 2.5 the level- n circles form a single $\mathrm{PSL}_2(\mathbb{Z})$ -set equivariantly isomorphic to the positive definite forms of discriminant $1 - n^2$, and the stabilizer of a circle equals the stabilizer of its hyperbolic center (an isometry fixing the center and preserving the arrangement fixes the circle, and conversely). It therefore suffices to prove:

Incidence lemma. *Let $z_0 \in \mathbb{H}$ with $s = |\mathrm{Stab}_{\mathrm{PSL}_2(\mathbb{Z})}(z_0)|$. Then the orbit of z_0 , counted in T with weight $\frac{1}{2}$ on the edges, has total weight $3/s$.*

The triangle T is a tile of the Farey tessellation, on which $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$ acts transitively with tile stabilizer of order 3 (generated by $\rho : z \mapsto 1/(1-z)$). Let $t(z) \in \{1, 2\}$ be the number of closed tiles containing z (1 interior, 2 on an edge; tessellation vertices are ideal, hence never in $\mathbb{H}$); t is constant on Γ -orbits, and the weight of $z \in \bar{T}$ is exactly $1/t(z)$. Count $\{\gamma \in \Gamma : \gamma z_0 \in \bar{T}\}$ in two ways: fibering over the point γz_0 gives $s \cdot \#(\Gamma z_0 \cap \bar{T})$; fibering over the tile $\gamma^{-1}T \ni z_0$ gives $3t(z_0)$. Hence the weighted count is $\#(\Gamma z_0 \cap \bar{T})/t(z_0) = 3/s$. Summing $3/s$ over classes gives $3H(n^2 - 1)$. $\square$

Remark 3.4 (which stabilizers occur). $s = 2$ would require the class of $f \cdot (1, 0, 1)$, i.e. $n^2 - 4f^2 = 1$: impossible for $n \geq 2$. So no level- n circle is centered at an order-2 elliptic point. $s = 3$ requires $n^2 - 3f^2 = 1$, the Pell equation with solutions $n = 2, 7, 26, 97, \dots$; for exactly these n one class has $s = 3$, whose representative in T is the circle centered at the centroid $\frac{1+i\sqrt{3}}{2}$. First cases: at $n = 2$, $3H(3) = 1$ and the unique circle is the inscribed circle of the ideal triangle — Schmidt’s original Farey circle, lying in $i\mathcal{S}$; at $n = 3$, $3H(8) = 3$, realized by six edge circles; at $n = 7$, $3H(48) = 10 = 18 \cdot \frac{1}{2} + 1$.

Machine verification: `alpha_circles.py` implements the level algorithm (solve $x^2 \equiv 1 - n^2 \pmod{4q}$ across q), draws the configurations (Figure 1), and confirms the weighted count $= 3H(n^2 - 1)$ for all $2 \leq n \leq 40$ and spot checks at $n = 301$ and $n = 1000$.

3.3. The spherical census: $4H(4(\ell^2 + 1))$. Project $\widehat{\mathbb{C}}$ stereographically onto the sphere of diameter 1 tangent to $\mathbb{C}$ at the origin (north pole $(0, 0, 1)$). On the sphere, lines are circles too, and compactness makes the census global: no fundamental domain is needed.

Proposition 3.5 (spherical radius). *Let ω be a Schmidt circle with Hermitian data $M = \begin{pmatrix} 2q & -\zeta \\ -\bar{\zeta} & 2m \end{pmatrix}$, $\det M = -1$. Its stereographic image is a circle*

Schmidt circles with $\alpha = 15$: 48 circles, weighted count $36 = 3H(224)$ (dashed = on an edge, weight $\frac{1}{2}$; dot = hyperbolic center; color = curvature)

Poincaré disk: ideal triangle at cube roots of unity, $\alpha = 15$

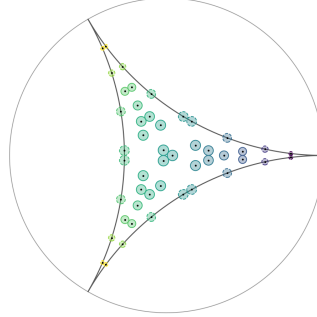


FIGURE 1. The 48 level-15 Schmidt circles with hyperbolic center in the ideal triangle $T = (0, 1, \infty)$, transported to the Poincaré disk, where the order-3 symmetry $z \mapsto 1/(1 - z)$ of T becomes exact threefold symmetry (the full symmetry is dihedral). Weighted count: $36 = 3H(224)$. Produced by `alpha_circles.py`.

of angular radius θ (the angle subtended at the sphere's center) with

$$\cot \theta = q + m =: \ell,$$

half the sum of curvature and co-curvature — an integer for every Schmidt circle. The lines $\operatorname{Im} z = k$ have $\ell = |k|$; the real line itself is the single great circle ($\ell = 0$).

Proof. A point (x_0, y_0, t) of the sphere satisfies $x_0^2 + y_0^2 + t^2 = 1$ and is the image of $z = (x_0 + iy_0)/(1 - t)$. Substituting into $2q|z|^2 - \zeta \bar{z} - \bar{\zeta} z + 2m = 0$ and clearing $(1 - t)$ shows the image lies in the plane $xx_0 + yy_0 + (m - q)t = m$ (with $\zeta = x + iy$). Its distance to the center $(0, 0, \frac{1}{2})$ is $\frac{|m+q|}{2} (|\zeta|^2 + (m - q)^2)^{-1/2}$, and $|\zeta|^2 + (m - q)^2 = 4qm + 1 + (m - q)^2 = 1 + (m + q)^2$; dividing by the radius $\frac{1}{2}$ gives $\cos \theta = |q + m|/\sqrt{1 + (q + m)^2}$. $\square$

Determinant -1 forces $qm \geq 0$, so after orientation $q, m \geq 0$ and $\ell \geq 0$: the spherical radii of Schmidt circles quantize to the integer-cotangent angles $\theta_\ell = \operatorname{arccot} \ell$, and no other radius occurs. We call ℓ the *spherical level*.

Theorem 3.6 (spherical census). *For $\ell \geq 1$, the number of Schmidt circles of spherical radius $\operatorname{arccot} \ell$ on the Riemann sphere is*

$$N_{\text{sph}}(\ell) = \frac{1}{3} r_3(\ell^2 + 1) = 4 H(4(\ell^2 + 1)),$$

where $r_3(n)$ is the number of representations of n as a sum of three squares. Moreover, the finite group $\Gamma_{\text{sph}} = \operatorname{SU}(2) \cap \operatorname{SL}_2(\mathbb{Z}[i])$ — in PSL_2 the Klein four-group $\{z, -z, 1/z, -1/z\}$ of rotations by π about the coordinate axes — acts freely on the level- ℓ circles, so the count on the quotient is exactly $H(4(\ell^2 + 1))$.

Proof. By Theorem 2.1, level- ℓ circles are the data (q, m, ζ) with $q + m = \ell$, $q, m \geq 0$, $\zeta = x + iy$, x even, y odd, $x^2 + y^2 = 4qm + 1$; the case $q = 0$ gives the lines $\text{Im } z = \pm \ell$, and every admissible datum occurs. Substituting $t := q - m$ (so $t \equiv \ell \pmod{2}$) turns the constraint into

$$x^2 + y^2 + t^2 = \ell^2 + 1, \quad x \equiv 0, \quad y \equiv 1, \quad t \equiv \ell \pmod{2},$$

a bijection (y odd excludes $x = y = 0$, so $|t| \leq \ell$ is automatic). For ℓ even, $\ell^2 + 1 \equiv 1 \pmod{4}$: every representation has exactly one odd coordinate, and permuting coordinates shows the three positions of the odd coordinate are equinumerous; our parity pattern selects one third. For ℓ odd, $\ell^2 + 1 \equiv 2 \pmod{4}$: exactly two odd coordinates, and the same argument applies to the position of the even one. Hence $N_{\text{sph}}(\ell) = r_3(\ell^2 + 1)/3$. Since $\ell^2 + 1 \equiv 1, 2 \pmod{4}$ (never $\equiv 0, 4, 7 \pmod{8}$), Gauss's theorem $r_3(n) = 12H(4n)$ applies [7, §4]. Freeness: the generators act by $(q, m, \zeta) \mapsto (q, m, -\zeta)$, $(m, q, \bar{\zeta})$, $(m, q, -\bar{\zeta})$; a fixed point would force ζ real (impossible: y odd) or $\zeta = yi$, $q = m$ with $y^2 = \ell^2 + 1$ (impossible for $\ell \geq 1$). $\square$

Remark 3.7. The antipodal map $z \mapsto -1/\bar{z}$ preserves the arrangement and pairs the level- ℓ circles freely. The three slices meet in exactly one point: $n^2 - 1 = 4(\ell^2 + 1)$ forces $(n, \ell) = (3, 1)$ — the discriminant -8 is seen twice by the arrangement, once hyperbolically at level 3 and once spherically at the 45° circles $\ell = 1$.

Machine verification: `spherical_moduli_invariants.py` (experiment A) verifies the census by exact enumeration for all $\ell \leq 20$ and the radius formula against fitted circumcircles of projected points to 40+ digits.

3.4. The triptych.

Theorem 3.8 (one arrangement, three geometries, three class-number counts).

The Gaussian Schmidt arrangement, graded by the natural radius invariant of each background geometry of $\widehat{\mathbb{C}}$, is counted by class numbers in all three:

<i>geometry</i>	<i>radius invariant</i>	<i>census</i>	<i>CM data of a level</i>
<i>Euclidean plane</i>	<i>curvature $2n$</i>	$N_e(n) = 2h(-4n^2)$	<i>conductor n over $\mathbb{Q}(i)$</i>
<i>hyperbolic plane</i>	<i>$\coth \rho = n$</i>	$3H(n^2 - 1)$	<i>discriminant $1 - n^2$</i>
<i>sphere</i>	<i>$\cot \theta = \ell$</i>	$4H(4(\ell^2 + 1))$	<i>discriminant $-4(\ell^2 + 1)$</i>

The Euclidean count is per unit area (Proposition 3.1, Theorem 7.6), the hyperbolic count is the weighted census of an ideal triangle (Theorem 3.3), and the spherical count is global (Theorem 3.6).

The three censuses read the coefficients of the classical weight-3/2 class-number generating functions [1, 22] along three quadratic progressions: $d = 4n^2$ (even squares), $d = n^2 - 1$ (shifted squares), $d = 4\ell^2 + 4$. The corresponding statements for Zagier's traces of singular moduli [23] appear below as Proposition 6.8 and its analogues.

4. THE INVOLUTION AND THE CLASS FORMULA

4.1. **What** $\sigma(X) = \bar{X}^{-1}$ **is.** Consider

$$\sigma : \mathrm{SL}_2(\mathbb{Z}[i]) \rightarrow \mathrm{SL}_2(\mathbb{Z}[i]), \quad \sigma(X) = \bar{X}^{-1}$$

(entrywise conjugation followed by inversion), an anti-automorphism of order two. Throughout, $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so $M_0 = iJ$, and for $X \in \mathrm{SL}_2$ one has $X^{-1} = J^{-1}X^\top J$.

Proposition 4.1. $\sigma(X) = J^{-1}X^\dagger J = M_0^{-1}X^\dagger M_0$. Thus σ is the adjoint anti-involution of the Hermitian form $h(u, v) = u^\dagger M_0 v$ attached to the real line $(h(Xu, v) = h(u, \sigma(X)v))$: in the language of central simple algebras, a unitary involution — an involution of the second kind — on $M_2(\mathbb{Q}(i))$, restricting to Galois conjugation on the center. Moreover

$$\mathrm{SU}(M_0; \mathbb{Z}[i]) = \mathrm{SL}_2(\mathbb{Z}) :$$

the unitary group of h over $\mathbb{Z}[i]$, the stabilizer of the oriented real line, and the fixed group of the Galois involution $\theta(X) = \bar{X} = \sigma(X)^{-1}$ all coincide with $\Gamma := \mathrm{SL}_2(\mathbb{Z})$.

Proof. The identity is immediate from $X^{-1} = J^{-1}X^\top J$ and $\bar{X}^\top = X^\dagger$. For the unitary group: $g^\dagger Jg = J$ together with $g^\top Jg = J$ (automatic for $\det g = 1$) forces $\bar{g} = g$. $\square$

Circles of the Schmidt arrangement correspond to cosets $X\Gamma$, and Γ -classes of circles to double cosets $\Gamma X\Gamma$; $(\mathrm{SL}_2(\mathbb{C}), \mathrm{SL}_2(\mathbb{R}))$ is the symmetric pair of the real structure θ , and σ is the Gelfand-trick anti-involution of that pair. Arithmetically it is far from trivial, as we now see.

Theorem 4.2 (Cartan embedding = circle inversion). *For $X \in \mathrm{SL}_2(\mathbb{Z}[i])$ write M_X for the Hermitian matrix of $X(\widehat{\mathbb{R}})$ as in (2.2). Then*

$$(4.1) \quad X\bar{X}^{-1} = -iJ\overline{M_X} = \begin{pmatrix} -iB & -iC \\ iA & i\bar{B} \end{pmatrix}, \quad M_X = \begin{pmatrix} A & B \\ \bar{B} & C \end{pmatrix}.$$

Consequently:

- (1) the Cartan image $Y = X\bar{X}^{-1}$ determines and is determined by the oriented circle $X(\widehat{\mathbb{R}})$; its fibers are the cosets $X\mathrm{SL}_2(\mathbb{Z})$;
- (2) $v \mapsto Y\bar{v}$ is the anti-holomorphic inversion in the circle $X(\widehat{\mathbb{R}})$, and $Y\bar{Y} = I$ (the Cartan image consists of twisted involutions);
- (3) $\mathrm{tr}(X\bar{X}^{-1}) = -2\alpha_\pm(X(\widehat{\mathbb{R}}))$, the level with a sign recording orientation;
- (4) for a level- n circle, the normalized product of the reflections of $\mathbb{H}^3$ in the hemispheres over $X(\widehat{\mathbb{R}})$ and over $\widehat{\mathbb{R}}$ is a hyperbolic translation of length $2\mathrm{arccosh} n$ along their common perpendicular, with characteristic polynomial $\lambda^2 + 2n\lambda + 1$; every level- n Schmidt circle thus determines a closed geodesic of length $2\mathrm{arccosh} n$ in the Bianchi orbifold $\mathrm{PSL}_2(\mathbb{Z}[i]) \backslash \mathbb{H}^3$, with real quadratic trace field $\mathbb{Q}(\sqrt{n^2 - 1})$.

Proof. (4.1) is a direct computation from (2.3) applied to X^{-1} , using $\bar{X}^{-1} = J^{-1}X^\dagger J$. (1) The right side of (4.1) is a bijective affine function of M_X ; and $M_{X\gamma} = M_X$ for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ while conversely $Y(X') = Y(X)$ forces $X^{-1}X' \in \mathrm{SL}_2(\mathbb{R}) \cap \mathrm{SL}_2(\mathbb{Z}[i]) = \mathrm{SL}_2(\mathbb{Z})$. (2) Rescaling Y by $1/(iA)$ puts it in the classical form $\begin{pmatrix} z_0 & r^2 - |z_0|^2 \\ 1 & -\bar{z}_0 \end{pmatrix}$ of the inversion in the circle of center z_0 , radius r ; conceptually, $\widehat{\mathbb{R}}$ is the fixed locus of $z \mapsto \bar{z}$, so $X(\widehat{\mathbb{R}})$ is the fixed locus of $X \circ \mathrm{conj} \circ X^{-1}$, whose Möbius part is Y . (3) $\mathrm{tr} Y = -iB + i\bar{B} = 2\mathrm{Im} B = -2y$ with y the oriented height numerator, which is $\alpha_\pm$ by Proposition 2.4. (4) The composite of the two inversions is the Möbius transformation $-Y$ -normalized, an isometry of $\mathbb{H}^3$ with characteristic polynomial $\lambda^2 + 2n\lambda + 1$ and eigenvalue $-(n + \sqrt{n^2 - 1})$; its translation length is $2\log(n + \sqrt{n^2 - 1}) = 2\mathrm{arccosh} n$. Both reflecting planes are $\mathrm{PSL}_2(\mathbb{Z}[i])$ -hemispheres, so the composite lies in $\mathrm{PSL}_2(\mathbb{Z}[i])$ and its axis projects to a closed geodesic. $\square$

Remark 4.3. Statement (4) is the three-dimensional, σ -twisted analogue of the trace- t conjugacy classes underlying reciprocal geodesics on the modular surface [17]: the length spectrum value $2\mathrm{arccosh} n$ arises here with multiplicity governed by $H(n^2 - 1)$ -data, through the fibers of the circle-to-geodesic map described in Corollary 4.10 below. We know of no prior appearance of this correspondence.

4.2. Fixed points: exactly three twisted classes. By Proposition 4.1 one has $\sigma(X) = X$ iff M_0X is Hermitian, iff $X = M_0M$ with M integral Hermitian of determinant -1 : *the fixed points of σ are precisely the matrices of circle inversions $z \mapsto M_0M(\bar{z})$, one for each integral Hermitian form of determinant -1 .* The natural action preserving the fixed set is the twisted conjugation $X \mapsto \bar{g}Xg^{-1}$, matching the action (2.2) on forms; classifying fixed points up to twisted conjugacy is a Galois-cohomology problem (over the field, Hilbert 90 trivializes it; over $\mathbb{Z}[i]$ it is a genus question). For the classical theory of binary Hermitian forms over imaginary quadratic rings, going back to Bianchi, see [4, Ch. 9].

Proposition 4.4 (computational). *Among the integral binary Hermitian forms of determinant -1 with $|A|, |C| \leq 20$, there are exactly three $\mathrm{SL}_2(\mathbb{Z}[i])$ -classes:*

- (1) *the class of M_0 : the Schmidt arrangement $\mathcal{S}$ ($B \equiv i \pmod{2}$; horizontal lines, even curvatures);*
- (2) *the class of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: the rotated family $i\mathcal{S}$ ($B \equiv 1 \pmod{2}$; vertical lines);*
- (3) *the class of $\mathrm{diag}(1, -1)$ (the unit circle): all circles of odd curvature — a family that is not part of the Schmidt arrangement at all.*

The three classes are pairwise distinct for all forms (the diagonal and off-diagonal parities mod 2 are invariants); the completeness of the list is verified by exhaustive orbit enumeration within the stated bounds by the script `involution_experiments.py`. Orientation-reversed seeds land in the same three classes.

Thus “the Schmidt arrangement” is one twisted Galois class of real structures on $\mathbb{P}^1$; the involution σ is the algebraic shadow of this descent picture.

4.3. The class formula. σ maps $\Gamma X \Gamma \mapsto \Gamma \sigma(X) \Gamma$ (well defined since $\sigma(\Gamma) = \Gamma$), hence induces an involution $\hat{\sigma}$ on Γ -classes of oriented Schmidt circles, i.e. — via Proposition 2.5, at odd level n — on form classes of discriminant $D = 1 - n^2$. Composition with the reflection $z \mapsto \bar{z}$ (which returns images to the upper half-plane and preserves each form class) makes $\hat{\sigma}$ an involution of the class set. Here is what it is.

Theorem 4.5 (class formula). *Let $n \geq 3$ be odd, $D = 1 - n^2$, and let f be a primitive positive definite form of discriminant D , with $X \in \mathrm{SL}_2(\mathbb{Z}[i])$ any matrix whose circle $X(\widehat{\mathbb{R}})$ lies in the $\mathrm{SL}_2(\mathbb{Z})$ -class of f . Then the circle of $\sigma(X)$ lies in the lower half-plane with reversed orientation and level $-n$, and its reflection into the upper half-plane lies in the class*

$$\hat{\sigma}[f] = [\mathfrak{r}_n][f]^{-1}, \quad \mathfrak{r}_n = \left[\left(\frac{n-1}{2}, 0, \frac{n+1}{2} \right) \right],$$

product and inverse taken in the form class group of discriminant D .

The twist is a canonical *ambiguous* (2-torsion) class: as an ideal, $\mathfrak{r} = (\frac{n-1}{2}, \omega)$ with $\omega = \sqrt{D}/2$, and $\mathfrak{r}^2 = (\frac{n-1}{2})$, while $\mathfrak{s} = (\frac{n+1}{2}, \omega)$ satisfies $[\mathfrak{s}] = [\mathfrak{r}]$ and $\mathfrak{r}\mathfrak{s} = (\omega)$. Since $[f]^{-1} = [\bar{\mathfrak{a}}]$ for the ideal class $[\mathfrak{a}]$ of f , the theorem reads

$$\hat{\sigma} : [\mathfrak{a}] \mapsto [\mathfrak{r}_n \bar{\mathfrak{a}}] :$$

Galois conjugation of the class group twisted by a distinguished ambiguous class — formally identical to the Atkin–Lehner action $w_N : [\mathfrak{a}] \mapsto [\mathfrak{n}\bar{\mathfrak{a}}]$ on Heegner points, with $\mathfrak{r}_n$ in the role of $\mathfrak{n}$. Geometrically, $\mathfrak{r}_n$ is the class of the circles tangent to the lines of the arrangement: its representative circle has curvature $n - 1$, center $\frac{ni}{n-1}$, and touches $\mathrm{Im} z = 1$.

The proof occupies the rest of this subsection. We use the ideal dictionary in the following normalization (Gauss; see [2, §7]): $K = \mathbb{Q}(\sqrt{D})$ with $\sqrt{D} := i\sqrt{n^2 - 1}$, $\omega = \sqrt{D}/2$, $\mathcal{O} = \mathbb{Z}[\omega]$ the order of discriminant D ; for a proper $\mathcal{O}$ -ideal $\mathfrak{b}$ with $\mathbb{Z}$ -basis (α, β) , the basis is *positively oriented* if $\mathrm{Im}(\bar{\alpha}\beta) > 0$, and $f_{\mathfrak{b}}(s, t) = N(s\alpha + t\beta)/N(\mathfrak{b})$; then $[\mathfrak{b}] \mapsto [f_{\mathfrak{b}}]$ is an isomorphism onto the primitive form class group, a negatively oriented basis producing the inverse class; $\mathfrak{a}_f := \mathbb{Z}a + \mathbb{Z}(\frac{b}{2} + \omega)$ satisfies $f_{\mathfrak{a}_f} = f$; and $\mathfrak{s} = \mathbb{Z}\frac{n+1}{2} + \mathbb{Z}\omega$ has $f_{\mathfrak{s}} = (\frac{n+1}{2}, 0, \frac{n-1}{2})$, so $[\mathfrak{s}] = \mathfrak{r}_n$, with $\bar{\mathfrak{s}} = \mathfrak{s}$, confirming that $\mathfrak{r}_n$ is 2-torsion (as it must be for $\hat{\sigma}$ to be an involution). A form $f = (a, b, c)$ of discriminant

D (b automatically even) corresponds to the circle

$$M_f = \begin{pmatrix} 2a & b - ni \\ b + ni & 2c \end{pmatrix} \quad (\text{curvature } 2a, \text{ center } \frac{-b+ni}{2a}).$$

Since $\hat{\sigma}$ is defined on double cosets, it suffices to prove the theorem for one X per class; we construct one in closed form.

Lemma 4.6 (explicit unitary basis; the closed-form section). *Let $f = (a, b, c)$ have discriminant $D = 1 - n^2$, and let*

$$\mathcal{K} := \{(s, t) \in \mathbb{Z}^2 : (n+1)s \equiv bt, \quad bs \equiv -(n-1)t \pmod{2a}\}.$$

Then $\mathcal{K}$ has index a in $\mathbb{Z}^2$. For any basis $w_1 = (s_1, t_1)$, $w_2 = (s_2, t_2)$ of $\mathcal{K}$ with $t_1 s_2 - s_1 t_2 = a$, put $u_k = s_k + it_k$ and

$$v_k = \frac{(n+bi)u_k + \bar{u}_k}{2ia} \quad (k = 1, 2).$$

Then

$$P = \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix} \in \text{SL}_2(\mathbb{Z}[i]) \quad \text{and} \quad P^\dagger M_0 P = M_f;$$

equivalently, $X := P^{-1}$ satisfies $X(\widehat{\mathbb{R}}) = \text{the circle } M_f$.

Proof. Index. $\mathcal{K} = \ker(R \bmod 2a)$ for $R = \begin{pmatrix} n+1 & -b \\ b & n-1 \end{pmatrix}$, $\det R = n^2 - 1 + b^2 = 4ac$. All entries of R are even and $\gcd(\frac{n+1}{2}, \frac{n-1}{2}) = 1$, so the Smith invariants of R are $(2, 2ac)$; hence $\#\ker(R \bmod 2a) = \gcd(2, 2a) \gcd(2ac, 2a) = 4a$ and $[\mathbb{Z}^2 : \mathcal{K}] = (2a)^2/4a = a$.

Integrality of v_k . $(n+bi)(s+it) + (s-it) = [(n+1)s - bt] + i[bs + (n-1)t]$, so $2ia \mid (n+bi)u_k + \bar{u}_k$ iff $(s_k, t_k) \in \mathcal{K}$.

Gram identities. Write $z := u_1 \bar{u}_2$, so $\text{Im } z = t_1 s_2 - s_1 t_2 = a$. Then $h(p_1, p_1) = 2 \text{Im}(u_1 \bar{u}_2) = 2a$; $\det P = (u_1 \bar{u}_2 - u_2 \bar{u}_1)/(2ia) = 1$; $\bar{u}_1 v_2 - \bar{u}_2 v_1 = (n+bi)(\bar{u}_1 u_2 - \bar{u}_2 u_1)/(2ia) = -(n+bi)$, giving $h(p_1, p_2) = b - ni$; and in $v_1 \bar{v}_2$ the cross terms are conjugate, hence real, so

$$\text{Im}(v_1 \bar{v}_2) = \frac{(n^2 + b^2) \text{Im } z + \text{Im } \bar{z}}{4a^2} = \frac{(n^2 + b^2 - 1)a}{4a^2} = \frac{4ac}{4a} = c,$$

i.e. $h(p_2, p_2) = 2c$. (Here p_1, p_2 denote the columns of P and h the form of Proposition 4.1.) $\square$

For example $n = 3$, $f = (1, 0, 2)$: $\mathcal{K} = \mathbb{Z}^2$, and $w_1 = (0, 1)$, $w_2 = (1, 0)$ give $P = \begin{pmatrix} i & 1 \\ 1 & -2i \end{pmatrix}$. Lemma 4.6 is of independent computational value: it writes down, in closed form, an element of $\text{SL}_2(\mathbb{Z}[i])$ realizing any prescribed Schmidt circle — previously available only through the descent algorithm of Theorem 2.1. It is the engine of most verification scripts in this paper, and of the phase computations of Section 6.

Lemma 4.7 (the Gram matrix of the σ -image). *Let $X = P^{-1}$, so $\sigma(X) = \bar{P}$, and let $N = (P^{-1})^\dagger M_0 P^{-1}$, so that the image circle is $M_{\sigma X} = -\bar{N}$. With $g(s, t) := (n+1)s^2 + (n-1)t^2$ and polarization $g(w, w') = (n+1)ss' + (n-1)tt'$:*

$$N = \begin{pmatrix} \frac{g(w_2)}{a} & -\frac{g(w_1, w_2)}{a} - ni \\ -\frac{g(w_1, w_2)}{a} + ni & \frac{g(w_1)}{a} \end{pmatrix},$$

and all displayed rational numbers are integers, the diagonal ones even.

Proof. The columns of P^{-1} are $(v_2, -u_2)^\top$ and $(-v_1, u_1)^\top$. Using $\frac{1}{2ia} = \frac{-i}{2a}$:

$$N_{11} = -2 \operatorname{Im}(v_2 \bar{u}_2) = \frac{\operatorname{Re}[(n+bi)|u_2|^2 + \bar{u}_2^2]}{a} = \frac{n|u_2|^2 + \operatorname{Re}(u_2^2)}{a} = \frac{g(w_2)}{a},$$

symmetrically $N_{22} = g(w_1)/a$, and $N_{12} = i(u_1 \bar{v}_2 - \bar{u}_2 v_1) = -g(w_1, w_2)/a - ni$ by the same expansion with $u_1 \bar{u}_2 = (s_1 s_2 + t_1 t_2) + ia$. *Integrality:* multiplying the two congruences defining $\mathcal{K}$ by s resp. t gives $(n+1)s^2 \equiv bst$ and $(n-1)t^2 \equiv -bst \pmod{2a}$, so $g(w) \equiv 0 \pmod{2a}$ on $\mathcal{K}$; polarizing, $2g(w_1, w_2) = g(w_1 + w_2) - g(w_1) - g(w_2) \equiv 0 \pmod{2a}$. $\square$

Two consequences are immediate. The oriented image $M_{\sigma X} = -\bar{N}$ has curvature entry $-g(w_2)/a < 0$: $\hat{\sigma}$ always reverses orientation. The un-oriented image lies in the lower half-plane with $|y| = n$: α is preserved. Reflecting by $z \mapsto \bar{z}$, the image form, read in the basis $(w_2, -w_1)$ of $\mathcal{K}$ (same orientation as (w_1, w_2)), is

$$(4.2) \quad f' := \left(\frac{g(w_2)}{2a}, -\frac{g(w_1, w_2)}{a}, \frac{g(w_1)}{2a} \right) = \frac{1}{2a} g|_{\mathcal{K}},$$

of discriminant $\operatorname{disc}(g) \cdot a^2 / (2a)^2 = D$. So the image class is the $\mathfrak{r}_n$ -form $\frac{1}{2}g = (\frac{n+1}{2}, 0, \frac{n-1}{2})$, restricted to the index- a lattice $\mathcal{K}$ and divided by a . It remains to recognize this as composition.

Lemma 4.8 ($\mathcal{K}$ is an ideal, and it factors). *Embed $\iota : \mathbb{Z}^2 \hookrightarrow K$, $\iota(s, t) = s \frac{n+1}{2} + t\omega$, so $\iota(\mathbb{Z}^2) = \mathfrak{s}$. Then $\mathfrak{c} := \iota(\mathcal{K})$ is an $\mathcal{O}$ -ideal, and for f primitive, $\mathfrak{c} = \mathfrak{s} \mathfrak{a}_f$.*

Proof. ω -stability. $\omega \iota(s, t) = \iota(\frac{1-n}{2}t, \frac{n+1}{2}s)$, and the image pair satisfies the $\mathcal{K}$ -congruences: substituting,

$$\begin{aligned} (n+1)\frac{1-n}{2}t - b\frac{n+1}{2}s &= -\frac{n+1}{2}[bs + (n-1)t], \\ b\frac{1-n}{2}t + (n-1)\frac{n+1}{2}s &= \frac{n-1}{2}[(n+1)s - bt], \end{aligned}$$

both multiples of the defining congruences, hence $\equiv 0 \pmod{2a}$.

Containment $\mathfrak{s} \mathfrak{a}_f \subseteq \mathfrak{c}$. The four generators of $\mathfrak{s} \mathfrak{a}_f$ have ι -coordinates $(a, 0)$, $(\frac{b}{2}, \frac{n+1}{2})$, $(0, a)$, $(\frac{1-n}{2}, \frac{b}{2})$, and each lies in $\mathcal{K}$: for the first and third use that b and $n \pm 1$ are even; for the second, $(n+1)\frac{b}{2} - b\frac{n+1}{2} = 0$ and $b\frac{b}{2} + (n-1)\frac{n+1}{2} = \frac{b^2 + n^2 - 1}{2} = 2ac \equiv 0 \pmod{2a}$; for the fourth, $(n+1)\frac{1-n}{2} - b\frac{b}{2} = -2ac$ and $b\frac{1-n}{2} + (n-1)\frac{b}{2} = 0$.

Equality. Both sides sit inside $\mathfrak{s}$ with the same index: $[\mathfrak{s} : \mathfrak{c}] = [\mathbb{Z}^2 : \mathcal{K}] = a$, and — here primitivity of f enters — $\mathfrak{a}_f$ is proper, so $[\mathfrak{s} : \mathfrak{s}\mathfrak{a}_f] = N(\mathfrak{a}_f) = a$. A containment of equal finite index is an equality. $\square$

Proof of Theorem 4.5. By Lemma 4.8 and $N \circ \iota = \frac{n+1}{2} \cdot \frac{1}{2}g$, the form (4.2) is the norm form of the proper ideal $\mathfrak{c} = \mathfrak{s}\mathfrak{a}_f$ of norm $\frac{n+1}{2}a$, computed in the basis $(\iota(w_2), \iota(-w_1))$. This basis is *negatively* oriented: for $w = (s, t)$, $w' = (s', t')$,

$$\operatorname{Im}(\overline{\iota(w)} \iota(w')) = \frac{n+1}{2} \cdot \frac{\sqrt{n^2-1}}{2} (st' - ts'),$$

and for the pair $(w_2, -w_1)$: $s_2(-t_1) - t_2(-s_1) = -a < 0$. A negatively oriented basis yields the opposite class, so by the dictionary,

$$[f'] = [f_{\mathfrak{c}}]^{-1} = ([\mathfrak{s}][\mathfrak{a}_f])^{-1} = [\mathfrak{r}_n]^{-1}[f]^{-1} = [\mathfrak{r}_n][f]^{-1},$$

the last step because $\mathfrak{r}_n$ is 2-torsion. $\square$

Note where each ingredient comes from: the *inversion* $[f]^{-1}$ comes from the orientation flip of complex conjugation inside σ (the $-a < 0$ computation); the *twist* $\mathfrak{r}_n$ comes from the Gram form $g(s, t) = n|u|^2 + \operatorname{Re}(u^2)$ — the norm form twisted by Galois conjugation $u \mapsto \bar{u}$, exactly the fingerprint σ leaves on the Gram matrix — whose class is forced to be the split of $n^2 - 1$ into the two even parts $\frac{n+1}{2} \cdot 2$.

Corollary 4.9. *$\hat{\sigma}$ has a fixed class iff $\mathfrak{r}_n$ is a square in the class group (iff $\mathfrak{r}_n$ lies in the principal genus); the fixed classes then form a coset of the 2-torsion. For $n = 11$ (discriminant -120 , class group $(\mathbb{Z}/2)^2$, three distinct nontrivial ambiguous classes — the decisive test case) there are none: $\hat{\sigma}$ sends the principal class to $[(5, 0, 6)] = \mathfrak{r}_{11}$, not to $[(2, 0, 15)]$ nor $[(3, 0, 10)]$, and swaps the other two, exactly as $[\mathfrak{r}][f]^{-1}$ predicts. For $n = 9$ (class group $\mathbb{Z}/4$, $\mathfrak{r}_9$ the square of a generator) the two order-4 classes $[(3, \pm 2, 7)]$ are fixed.*

Corollary 4.10 (fibers of the circle-to-geodesic map). *$\hat{\sigma}$ preserves each $\operatorname{SL}_2(\mathbb{Z}[i])$ -conjugacy class of Cartan images Y ; hence the unordered pair $\{[f], [\mathfrak{r}_n][f]^{-1}\}$ is an invariant of the closed geodesic of Theorem 4.2(4) attached to the circle, and the circle-to-geodesic map identifies exactly the $\hat{\sigma}$ -orbits of classes.*

Remark 4.11 (imprimitive classes; verified, unproved). On the imprimitive classes of content g , machine computation (`involution_classmap.py`, all classes of all odd $n \leq 41$) shows that $\hat{\sigma}$ is likewise $t_0 \cdot [f_0]^{-1}$ on the underlying primitive classes of discriminant $(1 - n^2)/g^2$, where the twist t_0 is an ambiguous class — in all computed cases the natural image of $\mathfrak{r}_n$ with the g^2 -part of its norm removed (e.g. $n = 15$, $g = 2$: $t_0 = [(2, 0, 7)]$). The only step of the proof using primitivity is the index count in Lemma 4.8; at the ramified prime 2 the ideal $\mathfrak{a}_f$ is not proper and $\mathfrak{s}\mathfrak{a}_f \subsetneq \mathfrak{c}$ can occur. A clean general statement on these strata remains open; we do not use them anywhere in this paper.

Machine verification: every identity of this section is checked in exact arithmetic — `involution_experiments.py` for (4.1) and the fixed-point characterization, `involution_classmap.py` for the class map at every class of every odd $n \leq 41$, and `proof_check.py` for every lemma of the proof (Lemmas 4.6–4.8 and the conclusion) at all 169 primitive classes with odd $n \leq 41$.

5. COMPOSITION IN CIRCLE LANGUAGE

Fix a level $n \geq 2$. By Proposition 2.5 the $\mathrm{SL}_2(\mathbb{Z})$ -classes of level- n circles carry the group structure of the form class group $\mathrm{Cl}(1 - n^2)$ (restricting to primitive classes). This section answers, in elementary terms: *what do the group operations do to the circles?* Think of $\omega_{q,x}$ as a fraction: a denominator q (half the curvature) and a numerator x (horizontal position in units of $\frac{1}{2q}$), the height locked to the level. Sliding by $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$ costs nothing (translations change x by $2q$), and the *identity class* is that of the top circle: $q = 1$, radius $\frac{1}{2}$, the largest circles of the level.

Proposition 5.1 (inverse = mirror). *The inverse of a circle class is its mirror image: the reflection $z \mapsto -\bar{z}$ sends $\omega_{q,x} \mapsto \omega_{q,-x}$, i.e. $f = (q, -x, m) \mapsto (q, x, m)$, the opposite form. Inversion in the unit circle sends $\omega_{q,x}$ to $\omega_{m,x}$ (curvature and co-curvature trade places), again the inverse class. In general, every orientation-reversing symmetry of the Schmidt arrangement inverts every class, being an improper equivalence of forms.*

Corollary 5.2 (2-torsion = mirror symmetry). *A circle class equals its own inverse iff some Γ -translate of the circle has its hyperbolic center on a mirror line of the modular tessellation. The three reduced shapes of ambiguous forms match the three basic mirrors:*

reduced form	circle	mirror
$b = 0$	$x = 0$: hyperbolic center on $\mathrm{Re} z = 0$	vertical line
$b = a$	$x = q$: center on $\mathrm{Re} z = \frac{1}{2}$	half-integer vertical
$a = c$	$q = m$: center on $ z = 1$	unit semicircle

(For $q = m$: $|z_{\mathrm{hyp}}|^2 = m/q$.) This makes the all-edge phenomenon of Section 3.2 transparent: at $n = 3, 5$ every class is ambiguous, so every circle of the ideal triangle sits on a mirror — and the triangle’s edges are mirrors.

Theorem 5.3 (composition = CRT = magnification). *To compose classes of $\omega_1 = \omega_{q_1, x_1}$ and $\omega_2 = \omega_{q_2, x_2}$ at level n :*

- (1) *slide the circles within their classes until q_1, q_2 are coprime (always possible: the half-curvatures occurring in a class are exactly the integers primitively represented by its form);*
- (2) *the composed class is that of the unique level- n circle with half-curvature $q_1 q_2$ whose numerator satisfies*

$$x_3 \equiv x_1 \pmod{2q_1}, \quad x_3 \equiv x_2 \pmod{2q_2}.$$

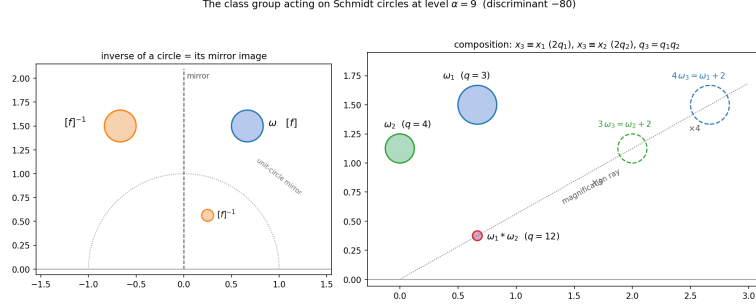


FIGURE 2. Inverse = mirror and composition = magnification at level $n = 9$. The three centers involved in each magnification are collinear with 0. Produced by `make_composition_figure.py`.

*Equivalently, geometrically: $\omega_1 * \omega_2$ is the unique arrangement circle of curvature $2q_1q_2$ that magnifies onto its factors: $q_2 \cdot (\omega_1 * \omega_2)$ is a translate of ω_1 and $q_1 \cdot (\omega_1 * \omega_2)$ is a translate of ω_2 . This is Gauss composition.*

Proof. Existence and uniqueness are CRT: the moduli $2q_1, 2q_2$ have gcd 2 and $x_1 \equiv x_2 \pmod{2}$ (both determined by the parity of n), so x_3 exists, unique mod $2q_1q_2$; the congruence $x_3^2 \equiv 1 - n^2 \pmod{4q_1q_2}$ follows from the two factors by CRT. The magnification formulation is one line: $x_3 \equiv x_1 \pmod{2q_1}$ says the q_2 -fold magnification of ω_3 (radius $\frac{1}{2q_1}$, center $\frac{x_3 + ni}{2q_1}$) differs from ω_1 by the integer translation $\frac{x_3 - x_1}{2q_1}$. (An integer magnification $z \mapsto kz$ with $k \mid q$ sends $\omega_{q,x}$ to a level- n arrangement circle of half-curvature q/k ; for $k \nmid q$ the image leaves the arrangement.) That the recipe is Gauss composition: in ideal language the circle $\omega_{q,x}$ corresponds to $\mathfrak{a} = \mathbb{Z}q + \mathbb{Z}\theta$, $\theta = \frac{-x + \sqrt{D}}{2}$, and for coprime q_1, q_2 and the CRT numerator x_3 ,

$$(q_1, \theta_3)(q_2, \theta_3) = (q_1q_2, q_1\theta_3, q_2\theta_3, \theta_3^2) = (q_1q_2, \theta_3),$$

since coprimality puts θ_3 in the product and $\theta_3^2 = -x_3\theta_3 - N(\theta_3)$ with $q_1q_2 \mid N(\theta_3)$. This is multiplication of ideals — equivalently Dirichlet composition of the concordant forms $(q_1, -x_3, q_2m_3)$ and $(q_2, -x_3, q_1m_3)$ [2, §3]. $\square$

Example 5.4 (level 9; $D = -80$, class group $\mathbb{Z}/4$). Compose $\omega_{3,4}$ (class $(3, 2, 7)$, a generator) with $\omega_{4,0}$ (class $(4, 0, 5)$): CRT gives $x_3 \equiv 4 \pmod{6}, \equiv 0 \pmod{8}$, so $x_3 = 16$ and the composite is $\omega_{12,16}$; indeed $4\omega_{12,16}$ has center $\frac{16+9i}{6}$, a translate of $\omega_{3,4}$, and in classes $[(3, 2, 7)] \cdot [(4, 0, 5)] = [(3, -2, 7)]$. Squaring $\omega_{3,4}$: the factors share $q = 3$, so slide one copy — $(3, 2, 7)$ also represents 7, giving $\omega_{7,2}$ — then $x_3 \equiv 4 \pmod{6}, \equiv 2 \pmod{14}$ gives $x_3 = 16$, composite $\omega_{21,16}$, whose form $(21, -16, 4)$ reduces to $(4, 0, 5)$: the order-2 class, as it must be.

5.1. Matrix-level composition. Suppose the circles are handed to us as $X(\widehat{\mathbb{R}}), Y(\widehat{\mathbb{R}})$ with $X, Y \in \mathrm{SL}_2(\mathbb{Z}[i])$. Three facts calibrate what the matrices buy.

Proposition 5.5. *Let X be oriented so that $\text{tr } Z_X = -2n$, $Z_X = X\bar{X}^{-1}$. Then*

$$W_X := \frac{i}{2}(Z_X + nI) = \begin{pmatrix} -x/2 & m \\ -q & x/2 \end{pmatrix}$$

is an integral matrix whose entries are the circle data of $X(\widehat{\mathbb{R}}) = \omega_{q,x}$ (curvature $2q$, numerator x , co-curvature $2m$): the form $f_X = (q, -x, m)$ can be read off X by one affine expression in the Cartan image, with no centers, radii or Bézout completions. Slides act by conjugation: $Z_{T^k X} = T^k Z_X T^{-k}$. Moreover W_X is the matrix of multiplication by $\omega = \sqrt{D}/2$ on the lattice $\Lambda = P\mathbb{Z}^2$ of Lemma 4.6: the integrality of W_X — equivalently the Schmidt congruence $\zeta \equiv i \pmod{2}$ — exhibits Λ as a module over the quadratic order $\mathcal{O} = \mathbb{Z}[W_X]$ of discriminant $1 - n^2$, and Cayley–Hamilton for W_X re-proves the trace identity $\text{tr}(X\bar{X}^{-1}) = \pm 2n$ intrinsically.

Proof. Immediate from (4.1): $Z_X = -iJ\bar{M}_X$ has entries carrying A, B, C , and the displayed affine shift extracts the real part of the form. For the module statement: on Λ the Hermitian form decomposes as $h|_\Lambda = 2f - ni\langle \cdot, \cdot \rangle$ with $\langle \cdot, \cdot \rangle$ the standard symplectic form; writing $Z = P\bar{P}^{-1}$, one finds $f(\lambda, \mu) = \langle \lambda, W\mu \rangle$ with $W = \frac{i}{4}(Z - Z^{-1})$, and $Z + Z^{-1} = (\text{tr } Z)I$ turns this into $W = \frac{i}{2}(Z + nI)$; then $W^2 = \frac{4 - (\text{tr } Z)^2}{16}I$ by Cayley–Hamilton while determinant bookkeeping forces $W^2 = \frac{D}{4}I$, so $\text{tr } Z = \pm 2n$. $\square$

Proposition 5.6 (align and conjugate). *Assume $\gcd(q_X, q_Y) = 1$. Slide both matrices, $X \mapsto T^{k_X} X$, $Y \mapsto T^{k_Y} Y$, so that both numerators equal the CRT value x_3 of Theorem 5.3 (the concordant position, in which the two centers are collinear with 0). Then with $E_k = \text{diag}(k, 1)$,*

$$W_3 = E_{q_Y}^{-1} W_X E_{q_Y} = E_{q_X}^{-1} W_Y E_{q_X} :$$

the two conjugates are equal as matrices, integral, and read off the composed circle $(q_X q_Y, x_3)$. An explicit $X_3 \in \text{SL}_2(\mathbb{Z}[i])$ with this circle is then given in closed form by Lemma 4.6.

Proof. Divisibility of the conjugates is exactly the CRT congruence, and their equality is the matrix form of “ ω_3 magnifies onto both factors”: conjugating W_{ω_3} by E_{q_Y} scales the anti-diagonal by $q_Y^{\pm 1}$, turning the data $(q_X q_Y, x_3, m_3)$ into $(q_X, x_3, q_Y m_3)$, the concordant representative of ω_1 ’s class, and symmetrically. $\square$

Remark 5.7 (no universal product formula). A plain matrix product does not compose: the circle of XY is $X(\omega_Y)$, the Möbius transport, generally at a completely different level, and its class is not determined by the two circles — inserting a middle representative ($XY \rightarrow X\gamma Y$, $\gamma \in \Gamma$, which changes neither circle) changes the answer. This is structural: composition is defined on classes, and some alignment of representatives — Gauss’s concordance — is unavoidable. What a product of Cartan matrices computes instead is the relative position of the pair:

$$\text{tr}(Z_X Z_Y^{-1}) = 2n^2 + 2x_1 x_2 - 4(q_1 m_2 + q_2 m_1),$$

a Γ -invariant of the two circles ($= 2$ iff they coincide); $Z_X Z_Y^{-1}$ is the composite of the two circle inversions, an isometry of $\mathbb{H}^3$ whose type encodes how the circles sit relative to each other.

Machine verification: `composition_check.py` verifies the mirror/inversion recipes and the CRT composite against Gauss composition on all 1913 ordered pairs of primitive classes at odd $n \leq 41$, and the mirror-shape characterization of 2-torsion on all 83 ambiguous classes; the script `matrix_composition_check.py` verifies the W_X read-off (93 matrices), the aligned conjugation composition (245 ordered pairs), the failure of naive products to be class-well-defined, and the trace-pairing identity.

6. THE PHASE INVARIANT

6.1. The sixth coordinate. Let $X \in \mathrm{SL}_2(\mathbb{C})$ with $\omega_1 = X(\widehat{\mathbb{R}})$ contained in the upper half-plane, and let $\omega_2 = \sigma(X)(\widehat{\mathbb{R}})$, which for integral Schmidt X in the positive-curvature normalization lies in the lower half-plane (Lemma 4.7). Write m_1 for the hyperbolic center of ω_1 and m_2 for the lower center of ω_2 (so $\bar{m}_2 \in \mathbb{H}$), $\alpha = \coth(\text{hyperbolic radius})$, and

$$\beta_1 = j(m_1), \quad \beta_2 = j(\bar{m}_2).$$

Both β_i are invariant under $X \mapsto \gamma X \gamma'$, $\gamma, \gamma' \in \mathrm{SL}_2(\mathbb{Z})$: left multiplication moves ω_1 by a Möbius map and fixes ω_2 ; right multiplication does the opposite; α is blind to both. The triple $(\alpha, \beta_1, \beta_2)$ has $1 + 2 + 2 = 5$ real dimensions against $\dim_{\mathbb{R}} \mathrm{SL}_2(\mathbb{C}) = 6$. The missing dimension is a phase, and the key to constructing it is an intertwining.

Proposition 6.1 (intertwining). *Let $Z = X \bar{X}^{-1}$ and $Z' = \bar{X}^{-1} X$. The fixed points of the Möbius transformation Z are exactly $\{m_1, \bar{m}_1\}$, those of Z' are $\{m_2, \bar{m}_2\}$, and $X Z' X^{-1} = Z$; consequently*

$$X \{m_2, \bar{m}_2\} = \{m_1, \bar{m}_1\}.$$

Proof. $v \mapsto Z\bar{v}$ is the inversion in ω_1 (Theorem 4.2), so $Z = \mathrm{inv}_{\omega_1} \circ \mathrm{conj}$, a composite of reflections of $\mathbb{H}^3$ in two disjoint geodesic planes — a loxodromic isometry of translation length $2 \operatorname{arccosh} \alpha$ whose axis endpoints are the unique pair symmetric with respect to both $\widehat{\mathbb{R}}$ and ω_1 , namely $\{m_1, \bar{m}_1\}$. The identity $X Z' X^{-1} = Z$ is immediate, and transports fixed points. $\square$

So X maps m_2 to m_1 or $\bar{m}_1$; integral Schmidt matrices in our normalization take the branch $X(m_2) = m_1$ (Lemma 6.5 below), and the derivative $X'(m_2) = (cm_2 + d)^{-2}$ is then a well-defined complex number. Its transformation under the two $\mathrm{SL}_2(\mathbb{Z})$ -actions is exactly cancelled by weight-2 factors:

Definition 6.2. With the branch $X(m_2) = m_1$,

$$\Theta(X) = \frac{j'(m_1) X'(m_2)}{j'(\bar{m}_2)};$$

on the branch $X(m_2) = \bar{m}_1$, replace $X'(m_2)$ by $\overline{X'(m_2)}$ and $j'(\bar{m}_2)$ by $j'(\bar{m}_2)$. More generally j' may be replaced by any *real kernel*: a meromorphic weight-2 function g for $\mathrm{SL}_2(\mathbb{Z})$ with q -expansion in $2\pi i \cdot \mathbb{R}[[q]][q^{-1}]$ (so that $g(\gamma z)\gamma'(z) = g(z)$ and $g(-\bar{\tau}) = -\overline{g(\tau)}$), finite and nonzero at the CM points involved. Excluded: $\alpha = 2$, where the centers are elliptic points of discriminant -3 and $j'(\rho) = 0$.

Proposition 6.3 (invariance and the fiber). *$\Theta(\gamma X \gamma') = \Theta(X)$ for all $\gamma, \gamma' \in \mathrm{SL}_2(\mathbb{Z})$. Moreover, fixing the five invariants, the identity component of the fiber through X is $X\{h_t\}$, where $h_t \subset \mathrm{SL}_2(\mathbb{R})$ is the elliptic one-parameter group fixing $\{m_2, \bar{m}_2\}$; along it $|\Theta(Xh_t)|$ is constant and $\arg \Theta(Xh_t)$ moves linearly at rate $\sqrt{\alpha^2 - 1}$. Hence $(\alpha, \beta_1, \beta_2, \arg \Theta)$ — six real dimensions — specify the double coset $\mathrm{SL}_2(\mathbb{Z}) X \mathrm{SL}_2(\mathbb{Z})$ up to finite ambiguity (finite j -fibers and stabilizers), on the region where the circles avoid the elliptic points.*

Proof. Left invariance: $\sigma(\gamma X) = \sigma(X)\gamma^{-1}$ fixes ω_2 and m_2 , while $m_1 \mapsto \gamma m_1$ and $(\gamma X)'(m_2) = \gamma'(m_1)X'(m_2)$; the weight-2 law $j'(\gamma m_1) = \gamma'(m_1)^{-1}j'(m_1)$ cancels the factor. Right invariance: m_1 is fixed, $m_2 \mapsto \gamma'^{-1}m_2$, and the derivative and denominator factors cancel the same way, using that γ' has real entries so conjugation commutes with it. Fiber: a real h preserves $\widehat{\mathbb{R}}$, hence fixes ω_1 -data exactly when it fixes ω_2 , i.e. $h \in \{h_t\}$; since $h_t(m_2) = m_2$, the chain rule gives $\Theta(Xh_t) = \Theta(X)h'_t(m_2)$ exactly, and the multiplier $h'_t(m_2)$ of an elliptic element at its fixed point has modulus one; its argument is linear in t with rate $\sqrt{\alpha^2 - 1}$ in the normalization of the loxodromic axis (machine-checked; the rate is the derivative of the rotation about the axis of Proposition 6.1). $\square$

6.2. Quantization on the Bianchi group, and the two laws. For level- n Schmidt matrices, define the *normalized phase*

$$u(X) := \varepsilon \Theta(X), \quad \varepsilon = n + \sqrt{n^2 - 1} = e^{\ell_{\mathrm{geod}}/2},$$

where $\ell_{\mathrm{geod}} = 2 \operatorname{arccosh} n$ is the geodesic length of Theorem 4.2(4); ε is the fundamental totally positive unit of norm 1 of the real quadratic order of discriminant $4(n^2 - 1)$. By Proposition 6.3, Θ is constant on classes and we write u_f . Throughout, $n \geq 3$ is odd, $N = n^2 - 1$, $D = 1 - n^2$, $f = (a, b, c)$ runs over primitive classes, and X is the canonical matrix of Lemma 4.6.

Lemma 6.4 (multiplier). *In the normalization $M_X = \begin{pmatrix} 2q & -\zeta \\ -\bar{\zeta} & 2m \end{pmatrix}$, $\zeta = x + ni$, the Cartan matrix $Z = X\bar{X}^{-1} = -iJ\overline{M_X}$ satisfies*

$$Z \begin{pmatrix} m_1 \\ 1 \end{pmatrix} = -\varepsilon \begin{pmatrix} m_1 \\ 1 \end{pmatrix}, \quad m_1 = \frac{x + i\sqrt{N}}{2q},$$

hence $Z'(m_1) = \varepsilon^{-2} = e^{-\ell_{\mathrm{geod}}}$; and symmetrically $Z' = \bar{X}^{-1}X$ has $Z' \begin{pmatrix} m_2 \\ 1 \end{pmatrix} = -\varepsilon \begin{pmatrix} m_2 \\ 1 \end{pmatrix}$ with m_2 the lower center.

Proof. The bottom row of Z is $(2iq, -i\bar{\zeta})$, and $2iqm_1 - i\bar{\zeta} = i(x + i\sqrt{N}) - i(x - ni) = -\sqrt{N} - n$; at a fixed point the pairing of the bottom row with $(m_1, 1)^\top$ is the eigenvalue, and the Möbius derivative there is (eigenvalue) $^{-2}$. $\square$

Lemma 6.5 (branches). $\bar{X}^{-1}(m_1) = m_2$, $X(m_2) = m_1$, and $X^{-1}(\bar{m}_1) = \bar{m}_2$.

Proof. From $Z'\bar{X}^{-1} = \bar{X}^{-1}Z$, the vector $\bar{X}^{-1}(m_1, 1)^\top$ is a $(-\varepsilon)$ -eigenvector of Z' ; the eigenvalues $-\varepsilon^{\pm 1}$ are distinct, so by Lemma 6.4 this eigenline is spanned by $(m_2, 1)^\top$: $\bar{X}^{-1}(m_1) = m_2$, i.e. $\bar{X}(m_2) = m_1$. Conjugating entries and argument gives $X(\bar{m}_2) = \bar{m}_1$, and since X permutes the pairs, $X(m_2) = m_1$; conjugating $\bar{X}^{-1}(m_1) = m_2$ gives the third identity. $\square$

Theorem 6.6 (functional equations). *Let g be any real kernel as in Definition 6.2 and $\Theta = \Theta_g$. With $X^* := R\bar{X}R$, $R = \text{diag}(-1, 1)$ (the mirror $z \mapsto -\bar{z}$ conjugation):*

$$(a) \quad \Theta(X^*) = \overline{\Theta(X)}; \quad (b) \quad \Theta(X) \overline{\Theta(X^{-1})} = \varepsilon^{-2} = e^{-\ell_{\text{geod}}}.$$

Consequently, for the normalized phase (any admissible kernel):

$$(\text{law 1}) \quad u_{f^{-1}} = \overline{u_f}; \quad (\text{law 2}) \quad u_f u_{\mathfrak{r}_n f} = 1.$$

In particular $u_f \in \mathbb{R}$ exactly on ambiguous classes; $|u_f| = 1$ on $\hat{\sigma}$ -fixed classes; $\prod_f u_f = \pm 1$; and at $n = 3$, $u = -1$ exactly.

Proof. (b) The circle of $Y := X^{-1}$ is $\bar{\omega}_2 \subset \mathbb{H}$ (since $X^{-1} = \overline{\sigma(X)}$ and entrywise conjugation conjugates the circle), so Y is again in our framework with $m_1(Y) = \bar{m}_2$, $m_2(Y) = \bar{m}_1$, branch “+” by Lemma 6.5. Then

$$\Theta(X) \overline{\Theta(Y)} = \frac{g(m_1)X'(m_2)}{g(\bar{m}_2)} \cdot \overline{\left(\frac{g(\bar{m}_2)Y'(\bar{m}_1)}{g(m_1)} \right)} = X'(m_2) \cdot \overline{(X^{-1})'(\bar{m}_1)} :$$

every kernel factor cancels. Now $\overline{(X^{-1})'(\bar{m}_1)} = (\bar{X}^{-1})'(m_1) = (\sigma X)'(m_1)$, and by Lemma 6.5 and the chain rule $X'(m_2)(\sigma X)'(m_1) = (X \circ \sigma X)'(m_1) = Z'(m_1) = \varepsilon^{-2}$ by Lemma 6.4. (a) $X^* = \nu X \nu$ with $\nu(z) = -\bar{z}$, so $m_i^* = -\bar{m}_i$, the branch stays “+”, and $X^{*'}(m_2^*) = \overline{X'(m_2)}$; applying $g(-\bar{\tau}) = -g(\tau)$ once in the numerator and once in the denominator, the two signs cancel. For the laws: the mirror inverts the circle class (Proposition 5.1) and $X \mapsto X^{-1}$ produces the class $[\mathfrak{r}_n][f]^{-1}$ (Theorem 4.5 applied to σ , plus conjugation); with $u = \varepsilon\Theta$, (a) gives law 1 and (b) with (a) give law 2. The listed consequences are formal; at $n = 3$ the class group is trivial and $\mathfrak{r}_3$ principal, so law 2 gives $u^2 = 1$, and $u = -1$ is the certified value (exact: Theorem 6.22). $\square$

Changing the kernel multiplies u_f by the explicit CM-algebraic factor $(g_2/g_1)(m_1)/\overline{(g_2/g_1)(\bar{m}_2)}$; both laws hold for every admissible kernel because the kernel cancelled in (b) and conjugated in (a). From now on $g = j'$.

6.3. Heegner pairs on a discriminant-coupled fiber product.

Proposition 6.7. *At level n , m_1 is the CM point of the class $[f]$, $\bar{m}_2 \in \mathbb{H}$ is the CM point of the class $\hat{\sigma}[f] = [\mathfrak{r}_n][f]^{-1}$, and $-m_2 \in \mathbb{H}$ is the CM point of the class $[\mathfrak{r}_n f]$; consequently $\beta_1 = j(\tau_{[f]})$ and $\beta_2 = j(\tau_{[\mathfrak{r}_n][f]^{-1}})$, with $\bar{\beta}_2 = j(-m_2) = j(\tau_{[\mathfrak{r}_n f]})$, are singular moduli of discriminant $D = 1 - n^2$, and over the primitive level- n classes*

$$\prod_f (x - \beta_1(f)) = H_D(x) \in \mathbb{Z}[x],$$

the ring class polynomial of discriminant D (imprimitive circles contribute the strata H_{D/g^2}).

Proof. The first claim is Proposition 2.5 for m_1 and Theorem 4.5 for $\bar{m}_2$; then $-m_2$, the mirror image of $\bar{m}_2$, is the point of the inverse class $[\mathfrak{r}_n f]$, and j of mirror points are complex conjugates. The product formula is the definition of the ring class polynomial, whose integrality is classical [2, §13]. $\square$

Proposition 6.8 (traces). *With weights $1/s$ at the stabilized classes, one has $\sum_{\text{classes}} (\beta_1 - 744)/s = t(n^2 - 1)$, Zagier's trace of singular moduli [23]; the arrangement reads the coefficients of the associated weight-3/2 form along the shifted squares $d = n^2 - 1$. (Computational: verified against Zagier's published values for $n \leq 8$; `moduli_invariants.py`, experiment E.)*

Theorem 6.9 (simultaneous modular equations). *Let f be a primitive class at odd level n , and let $\Phi_m(x, y)$ denote the classical modular polynomial. Then*

$$\Phi_{\frac{n-1}{2}}(\beta_2, \bar{\beta}_1) = 0 \quad \text{and} \quad \Phi_{\frac{n+1}{2}}(\beta_2, \bar{\beta}_1) = 0.$$

The two degrees multiply to $\frac{n^2-1}{4} = |D|/4$: each $\hat{\sigma}$ -pair of Schmidt circles defines a point on the fiber product $X_0(\frac{n-1}{2}) \times_{X(1)} X_0(\frac{n+1}{2})$ — a Heegner-type configuration in which the level pair is coupled to the discriminant.

Proof. $\mathfrak{a}_{f_2} = \mathfrak{r}_n \bar{\mathfrak{a}}_f$ with $N(\mathfrak{r}_n) = \frac{n-1}{2}$, so the elliptic curve with $j = \beta_2$ is cyclically $\frac{n-1}{2}$ -isogenous to the curve with $j = \bar{\beta}_1$ (the kernel is the $\mathfrak{r}_n$ -torsion, cyclic because $\mathcal{O}/\mathfrak{r}_n \cong \mathbb{Z}/\frac{n-1}{2}$); using the partner ideal $\mathfrak{s}$ of norm $\frac{n+1}{2}$ in the same class, it is also $\frac{n+1}{2}$ -isogenous to it. Both statements are the defining property of Φ_m [2, §11]. $\square$

6.4. The algebraic closed form and the Norm Lemma. Write h_2 for the weight-2 homogeneous lattice function with $h_2([\tau, 1]) = C \cdot F(\tau)$, $F = E_4^2 E_6 / \Delta$, one universal constant C ($h_2(\lambda\Lambda) = \lambda^{-2} h_2(\Lambda)$); recall $j' = -2\pi i F$. Let $\mathfrak{b}_1 = [1, m_1]$ and $\mathfrak{b}_2 = [1, -m_2]$, proper fractional $\mathcal{O}$ -ideals, and let $\mu := c m_2 + d$ (bottom row of the canonical X), an element of the CM field $B := K(i) = \mathbb{Q}(i, \sqrt{n^2 - 1})$ containing the real quadratic $\mathbb{Q}(\sqrt{n^2 - 1})$.

Proposition 6.10 (closed form). $u_f = -\varepsilon \frac{h_2(\mu \mathbf{b}_1)}{h_2(\mathbf{b}_2)} = -\varepsilon \mu^{-2} \frac{h_2(\mathbf{b}_1)}{h_2(\mathbf{b}_2)}$.

In particular, by Shimura's algebraicity theorem for weight-2 ratios at CM points [19], u_f is algebraic and nonzero.

Proof. With real q -coefficients, $\overline{j'(\bar{m}_2)} = -j'(-m_2)$, and $-m_2 \in \mathbb{H}$ is the CM point of $[\mathfrak{r}_n f]$; homogeneity converts $j'(m_1)/j'(-m_2)$ into the h_2 -ratio of the two lattices, and $X'(m_2) = \mu^{-2}$. Nonvanishing: j' vanishes only at the orbits of i and ρ , of discriminants $-4, -3 \neq D$ for $n \geq 3$; and $\mu \neq 0$. $\square$

Lemma 6.11 (Norm Lemma). *Let q_1, q_2 be the half-curvatures of the two circles and τ the generator of $\text{Gal}(B/\mathbb{Q}(i))$ (so $\tau|_K$ is the conjugation of K and $\varepsilon^\tau = \varepsilon^{-1}$). Then*

$$|\mu|^2 = \varepsilon \frac{q_1}{q_2}, \quad |\mu^\tau|^2 = \frac{q_1}{\varepsilon q_2} :$$

the unit ε is exactly the discrepancy between the two archimedean places of μ . Moreover, as ideals of $\mathcal{O}_B$,

$$(\mu) \mathcal{O}_B = \mathfrak{m}_2 \mathfrak{m}_1^{-1}, \quad \mathfrak{m}_i := \mathcal{O}_B + \mathcal{O}_B m_i.$$

Proof. Apply the Hermitian form $h(v, v) = v^\dagger M_0 v$ to $X(m_2, 1)^\top = \mu(m_1, 1)^\top$, using $h(Xv, Xv) = v^\dagger M_{X^{-1}} v$ with $M_{X^{-1}}$ the positive-curvature matrix of $\bar{\omega}_2$. For a level- n circle $M = (2q, -\zeta, 2m)$ with hyperbolic center z_h one computes $M(z_h) = -\sqrt{N}/(\varepsilon q)$ and $M(\bar{z}_h) = \varepsilon \sqrt{N}/q$; since $h((z, 1)^\top) = 2 \text{Im } z$, the left side is $|\mu|^2 \sqrt{N}/q_1$ and the right side $\varepsilon \sqrt{N}/q_2$. The τ -statement is the same computation on the conjugated intertwining $X(\bar{m}_2, 1)^\top = \mu^\tau(\bar{m}_1, 1)^\top$ (Lemma 6.5), where the argument is the hyperbolic center itself. The ideal identity: with $ev(s, t) := sm_2 + t$, we have $\mu \cdot 1 = ev(\text{row}_2 X)$ and $\mu m_1 = ev(\text{row}_1 X)$; since $\det X = 1$, the rows of X are a $\mathbb{Z}[i]$ -basis of $\mathbb{Z}[i]^2$ and ev is $\mathbb{Z}[i]$ -linear, so the $\mathcal{O}_B$ -span of $\mu \mathbf{b}_1$ equals that of $ev(\mathbb{Z}[i]^2) = \mathbb{Z}[i] + \mathbb{Z}[i]m_2$, i.e. $(\mu)\mathfrak{m}_1 = \mathfrak{m}_2$. (Taking norms recovers $N_{B/\mathbb{Q}}(\mu) = (q_1/q_2)^2$, consistent with the archimedean statement.) $\square$

6.5. First-power descent: the phase is the derivative of the modular correspondence. Set $r_0 = \frac{n-1}{2}$, $s_0 = \frac{n+1}{2}$, $\omega_0 = \sqrt{D}/2$, and recall $\mathfrak{r} = [r_0, \omega_0]$, $\mathfrak{s} = [s_0, \omega_0]$ ($N\mathfrak{r} = r_0$, $N\mathfrak{s} = s_0$, $[\mathfrak{s}] = [\mathfrak{r}]$, $\mathfrak{r}^2 = (r_0)$, $\mathfrak{r}\mathfrak{s} = (\omega_0)$).

Definition 6.12. For a proper $\mathcal{O}$ -ideal $\mathfrak{a}$, set $V(\mathfrak{a}) := h_2(\mathfrak{a})/h_2(\mathfrak{r}^{-1}\mathfrak{a})$. Weight-2 homogeneity kills scalars, so V descends to classes; write $V_f := V(\mathbf{b}_1)$ (the second lattice has class $[\mathfrak{r}f]$).

Lemma 6.13 (uniformization). *There is $\tau_0 \in \mathbb{H}$ with $\mathfrak{a} = \beta[\tau_0, 1]$ and $\mathfrak{r}^{-1}\mathfrak{a} = \beta\left[\tau_0, \frac{1}{\tau_0}\right]$ for one common β ; consequently*

$$V(\mathfrak{a}) = \frac{1}{r_0^2} \frac{F(\tau_0)}{F(r_0\tau_0)} = \frac{1}{r_0^2} \frac{j'(\tau_0)}{j'(r_0\tau_0)},$$

$$(j(\tau_0), j(r_0\tau_0)) = (\beta_1, \beta_2) := (j(\mathfrak{a}), j(\mathfrak{r}^{-1}\mathfrak{a})).$$

Proof. $\mathcal{O}/\mathfrak{r} \cong \mathbb{Z}/r_0$, so $\mathfrak{r}^{-1}\mathfrak{a}/\mathfrak{a}$ is cyclic of order r_0 ; pick a $\mathbb{Z}$ -basis adapted to the elementary divisors $(1, r_0)$ and orient. Then $\left[\tau_0, \frac{1}{r_0}\right] = \frac{1}{r_0} [r_0\tau_0, 1]$ gives the displayed value; the constants cancel in the ratio. $\square$

Lemma 6.14 (the pair is a smooth point). *At every primitive class and every odd $n \geq 3$, the point (β_1, β_2) is a smooth point of the affine curve $\Phi_{r_0} = 0$, and $\Phi_x(\beta_1, \beta_2) \neq 0 \neq \Phi_y(\beta_1, \beta_2)$.*

Proof. Branches of $\{\Phi_{r_0} = 0\}$ through (β_1, β_2) correspond to cyclic r_0 -isogenies $E_{\mathfrak{a}} \rightarrow E'$ with $j(E') = \beta_2$, modulo automorphisms. Since j determines the lattice up to homothety and singular moduli of distinct orders are distinct, every such isogeny has kernel $E[\mathfrak{c}']$ for an *invertible* $\mathcal{O}$ -ideal $\mathfrak{c}'$ of norm r_0 with $[\mathfrak{c}'] = [\mathfrak{r}]$. Such ideals correspond to primitive representations of r_0 by the form $(r_0, 0, s_0)$ up to automorphs; since $s_0 > r_0$, the only representation is $r_0 = r_0 \cdot 1^2$: the branch is unique. It is smooth with nonvanishing partials because its parametrization $\tau \mapsto (j(\tau), j(r_0\tau))$ near τ_0 has $j'(\tau_0) \neq 0 \neq j'(r_0\tau_0)$ (both points are CM of discriminant $D \notin \{-3, -4\}$, both lattices proper), and a unique smooth branch is a smooth point of the reduced irreducible curve; finally $\Phi_x j'(\tau_0) + \Phi_y r_0 j'(r_0\tau_0) = 0$ shows the partials vanish together if at all. $\square$

Proposition 6.15 (correspondence derivative). $V(\mathfrak{a}) = -\frac{1}{r_0} \cdot \frac{\Phi_y(\beta_1, \beta_2)}{\Phi_x(\beta_1, \beta_2)}$, $\Phi = \Phi_{r_0}$. *In particular $V(\mathfrak{a}) \in \mathbb{Q}(\beta_1, \beta_2) \subseteq H$, the ring class field: the weight-2 ratio is a rational expression in two singular moduli.*

Proof. Differentiate $\Phi(j(\tau), j(r_0\tau)) \equiv 0$ at τ_0 and use Lemma 6.13 and $\Phi_x \neq 0$. $\square$

Proposition 6.16 (reciprocity at first power). *For $\sigma \in \text{Gal}(\bar{\mathbb{Q}}/K)$ with Artin class $\mathfrak{c}(\sigma) \in \text{Cl}(\mathcal{O})$ on H : $\sigma(V(\mathfrak{a})) = V(\mathfrak{c}(\sigma)^{-1}\mathfrak{a})$. Complex conjugation gives $\iota(V(\mathfrak{a})) = V(\bar{\mathfrak{a}})$. Hence on classes, for every $\sigma \in \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$,*

$$\sigma(V_f) = V_{f^{e(\sigma)}\mathfrak{c}(\sigma)},$$

where $e(\sigma) = +1$ if $\sigma|_K = \text{id}$ and -1 otherwise, and $\mathfrak{c}(\sigma)$ is the Artin class of σ (resp. of $\sigma\iota$) on the ring class field.

Proof. Apply σ to the identity of Proposition 6.15. Since $\Phi \in \mathbb{Z}[x, y]$, $\sigma(V(\mathfrak{a})) = -\frac{1}{r_0} \Phi_y(\sigma\beta_1, \sigma\beta_2)/\Phi_x(\sigma\beta_1, \sigma\beta_2)$, and by the reciprocity law for ring class fields of orders [2, Thm. 11.36 and §9], $\sigma\beta_1 = j(\mathfrak{c}^{-1}\mathfrak{a})$ and $\sigma\beta_2 = j(\mathfrak{r}^{-1}(\mathfrak{c}^{-1}\mathfrak{a}))$: the image pair is the canonical pair of the translated class, again a smooth point (Lemma 6.14) evaluated by the same rational formula. For ι : Φ has integer coefficients, $\iota(j(\mathfrak{a})) = j(\bar{\mathfrak{a}})$, and $\bar{\mathfrak{r}} = \mathfrak{r}$. The dihedral composition rule is then formal. $\square$

Note what did *not* enter: no Kronecker limit formula, no elliptic units, no Siegel functions [11], no level structure beyond the j -line — only $\sigma(j(\mathfrak{a})) = j(\mathfrak{c}^{-1}\mathfrak{a})$. It remains to connect V_f to u_f .

Since $[\mathbf{b}_2] = [\mathfrak{r}][\mathbf{b}_1] = [\mathfrak{r}^{-1}\mathbf{b}_1]$, there is $\nu = \nu_f \in K^\times$, unique up to sign, with $\mathbf{b}_2 = \nu \mathfrak{r}^{-1}\mathbf{b}_1$. Homogeneity turns Proposition 6.10 into

$$(6.1) \quad u_f = -\varepsilon \mu^{-2} \nu^2 V_f = -r_0 \omega_f V_f, \quad \omega_f := \frac{\varepsilon \nu_f^2}{r_0 \mu_f^2},$$

a well-defined class quantity (only ν^2 enters, and u_f, V_f are class functions).

Lemma 6.17 (unit rigidity). $\omega_f \in \mu(B)$, the roots of unity of the CM field B .

Proof. Archimedean: the Norm Lemma gives $|\mu|^2 = \varepsilon q_1/q_2$ and $|\mu^\tau|^2 = q_1/(\varepsilon q_2)$, while $N(\nu) = N\mathbf{b}_2 \cdot N\mathfrak{r}/N\mathbf{b}_1 = r_0 q_1/q_2$ and $|\nu^\tau| = |\bar{\nu}| = |\nu|$; so $|\omega_f| = |\omega_f^\tau| = 1$ at both archimedean places. Ideal: $(\mu)\mathcal{O}_B = \mathfrak{m}_2 \mathfrak{m}_1^{-1}$ (Lemma 6.11) and $(\nu)\mathcal{O}_B = \mathfrak{m}_2 \mathfrak{m}_1^{-1}(\mathfrak{r}\mathcal{O}_B)$ from the definition of ν , while $\mathfrak{r}^2 = (r_0)$ and ε is a unit, so $(\omega_f) = \mathcal{O}_B$. All conjugates of the algebraic integer ω_f lie on the unit circle (B is CM), so Kronecker's theorem applies. $\square$

Theorem 6.18 ($\omega \equiv 1$). $\omega_f = 1$ for every primitive class f and every odd $n \geq 3$. Equivalently:

$$u_f = -r_0 V_f = \frac{\Phi_y(\beta_1, \beta_2)}{\Phi_x(\beta_1, \beta_2)}, \quad \Phi = \Phi_{r_0}, \quad \beta_1 = j(\mathbf{b}_1), \quad \beta_2 = j(\mathfrak{r}^{-1}\mathbf{b}_1).$$

Proof. By Lemma 6.17 and $\varpi\bar{\varpi} = \varepsilon/r_0$ for $\varpi := \mu/\nu$ (Norm Lemma), we have $\omega_f = \bar{\varpi}/\varpi$, so $\omega_f = 1$ iff μ/ν is real. Compute in the Hermite normal form of the basis of Lemma 4.6, which has $s_2 = 0$ (the second basis vector spans $\mathcal{K} \cap (0 \times \mathbb{Z})$). Three exact coordinates (each verified in rational arithmetic at every primitive class, odd $n \leq 21$; `first_power_descent.py`): writing $\varsigma(s, t) := ss_0 + t\omega_0$ and $\sigma_k := \varsigma(w_k)$, one has (i) $N \circ \varsigma = \frac{s_0}{2}g$ with g the Gram form of Lemma 4.7; (ii) $m_2 = \bar{\sigma}_1/\bar{\sigma}_2$; (iii) $\nu = \pm a \omega_0/\bar{\sigma}_2$. [(i) is one line; (ii) holds because by (i) the σ -circle Gram of Lemma 4.7 is the normalized norm form of the lattice $[\sigma_2, \sigma_1] = \mathfrak{s}\mathfrak{a}_f$, whose lower root is $\bar{\sigma}_1/\bar{\sigma}_2$ by the orientation of Lemma 4.6; (iii) follows from $\mathbf{b}_2 = [\sigma_2, \sigma_1]/\bar{\sigma}_2 = \mathfrak{s}\bar{\mathfrak{a}}_f/\bar{\sigma}_2$ with $\mathbf{b}_1 = \bar{\mathfrak{a}}_f/a$ and $\mathfrak{s} = (\omega_0)\mathfrak{r}^{-1}$.] With $s_2 = 0$: $\sigma_2 = t_2\omega_0$ is purely imaginary, so $\nu = \pm a/t_2 \in \mathbb{Q}$ by (iii); and by (ii),

$$u_2 m_2 = it_2 \cdot \frac{\bar{\sigma}_1}{-t_2 \omega_0} = -\frac{2\bar{\sigma}_1}{\sqrt{N}} \implies \mu = u_1 - u_2 m_2 = s_1 + it_1 + \frac{2(s_1 s_0 - it_1 \sqrt{N}/2)}{\sqrt{N}} = s_1 \frac{\sqrt{N} + 2s_0}{\sqrt{N}} :$$

the imaginary part cancels identically and μ is real. Hence ϖ is real and $\omega_f = 1$; explicitly $(\sqrt{N} + 2s_0)^2 = 4s_0\varepsilon$ gives $\mu^2 = \varepsilon s_1^2/r_0$ and $\omega_f = a^2/(s_1 t_2)^2 = 1$, since the orientation $t_1 s_2 - s_1 t_2 = a$ with $s_2 = 0$ forces $s_1 t_2 = -a$. $\square$

Theorem 6.19 (first-power dihedral law). For every $\sigma \in \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ and every primitive class f at every odd level $n \geq 3$:

$$\sigma(u_f) = u_{f^{e(\sigma)} \mathfrak{c}(\sigma)}.$$

Consequently:

- (1) u_f lies in the ring class field H of discriminant $1 - n^2$, with $\mathbb{Q}(u_f)$ cut out by the stabilizer $\{\sigma : f^{e(\sigma)} \mathbf{c}(\sigma) = f\}$;
- (2) the multiset $\{u_f\}_{f \in \text{Cl}}$ is $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ -stable and forms a single Galois orbit (translations act transitively and every class is an Artin class);
- (3) the pair-sums $S_x = u_f + u_f^{-1}$, $x = \{f, \mathbf{r}f\}$, satisfy $\sigma(S_x) = S_{x^{e(\sigma)} \mathbf{c}(\sigma)}$, and their fields are given by genus theory: S_x lies in the real part of the fixed field of $\text{Art}(\mathbf{r})$ whenever $x^2 \in \langle \mathbf{r} \rangle$;
- (4) the corresponding equivariance holds at every power; in particular no root-of-unity cocycle appears at any power of u_f .

Proof. $u_f = -r_0 V_f$ (Theorem 6.18) with $r_0 \in \mathbb{Q}$, and V obeys the dihedral law (Proposition 6.16). The consequences are formal; (3) uses that the reachable translations are $\text{Gal}(H/H \cap B)$ and the mirror/inversion mechanisms of Theorem 6.6. $\square$

Remark 6.20. Historically the Galois law was first proved for u_f^{12} by a Shimura–Siegel reciprocity argument with a root-of-unity cocycle controlled through a cocycle-ideal computation, and the descent to first power appeared to require the Kubert–Lang theory of modular units [11]. Theorem 6.18 dissolves the cocycle at the source: the sixth root that plagued the Δ -quotient expression of u^6 is an artifact of factoring h_2^6 , while the ratio h_2/h_2 itself is already a rational function on $X_0(r_0)$. Moduli-theoretically, $-\Phi_x/(r_0 \Phi_y) = dj_2/dj_1$ along $X_0(r_0)$: the phase of a Schmidt circle is the derivative of the modular correspondence at the Heegner point the circle defines. Law 2 ($u_{\mathbf{r}f} u_f = 1$) becomes the chain rule for the two branches through the point, and the partner ideal $\mathfrak{s}$ gives $V_f/V'_f = -s_0/r_0$ from $\mathbf{r}\mathfrak{s} = (\omega_0)$.

6.6. One level, one integer polynomial.

Theorem 6.21 (structure of the level polynomial). *Fix an odd $n \geq 3$ and set $\Pi_n(x) = \prod_{f \in \text{Cl}(1-n^2)} (x - u_f)$. Then $\Pi_n \in \mathbb{Q}[x]$, and $\Pi_n = m^k$ with m irreducible over $\mathbb{Q}$ and $k = h/\deg m$; the coincidence relation $u_f = u_g$ is translation-invariant, so it is the coset relation of a subgroup $J \leq \text{Cl}$ with $k = |J|$, and*

$$\Pi_n \text{ is irreducible} \iff \text{the } u_f \text{ are pairwise distinct} \iff J = 1.$$

Moreover Π_n is self-reciprocal, with $\prod_f u_f = 1$ for $n \geq 5$, and its coefficients are real; the primitive integer form $Q_n = d'_n \Pi_n \in \mathbb{Z}[x]$ is palindromic with positive leading coefficient d'_n , and $\mathbf{r} \notin J$ for $n \geq 5$ (else $|u_f| \equiv 1$, contradicting the exponential growth of the principal-class phase).

Proof. Rationality: the coefficients are algebraic (Proposition 6.10) and fixed by every σ , since by Theorem 6.19 each σ permutes the multiset $\{u_f\}$ through the bijection $f \mapsto f^{e(\sigma)} \mathbf{c}(\sigma)$. Transitivity (Theorem 6.19(2)) makes all irreducible factors equal. Applying a translation to an equality $u_f = u_g$ shows the relation is translation-invariant, so its classes are the cosets of

a subgroup J ; $J = 1$ iff all values are distinct. Palindromy is law 2 (the bijection $f \mapsto \mathfrak{r}f$ inverts every root, and for $n \geq 5$ the reduced form $\mathfrak{r}_n$ has $a \neq 1$, so $[\mathfrak{r}_n] \neq 1$, h is even and the roots cancel in pairs to give $\prod u_f = 1$); real coefficients are law 1. For $\mathfrak{r} \notin J$: $u_{\mathfrak{r}f} = u_f$ would give $u_f^2 = 1$ for all f by law 2, while $|u_1| = r_0^2 |F(\tau_1)/F(r_0\tau_1)| > 1$ grows exponentially (the principal CM point $\tau_1 = \frac{\sqrt{N}}{2}i$ has the largest imaginary part; crude q -expansion bounds give $\log |u_1| = \pi\sqrt{n^2 - 1}(1 + o(1))$). $\square$

Theorem 6.22 (computed levels; exact arithmetic). *For $n = 3, 5, 7, 9, 11, 13$ the polynomial Π_n was computed exactly — with no numerical identification anywhere — by the following route: (i) $H_D \in \mathbb{Z}[t]$ is certified by integer rounding with error $< 10^{-40}$; (ii) in $F_1 = \mathbb{Q}[t]/(H_D)$, the pairing root $\beta_2(t)$ is the root of $\gcd(H_D(y), \Phi_{r_0}(t, y))$, which has degree 1 (the exact-arithmetic reflection of Lemma 6.14); (iii) $u(t) = \Phi_y(t, \beta_2(t))/\Phi_x(t, \beta_2(t)) \in F_1$, and Π_n is the characteristic polynomial of multiplication by $u(t)$ on F_1 ; the modular polynomials $\Phi_1, \dots, \Phi_6$ are themselves constructed exactly from integer q -expansions. The resulting primitive integer polynomials are:*

$$\begin{aligned}
Q_3 &= x + 1 \\
Q_5 &= 6647x^2 + 30594194x + 6647 \\
Q_7 &= 11891x^2 + 80674200806x + 11891 \\
Q_9 &= 10565574794063311x^4 + 73919532109765731422845124x^3 \\
&\quad - 118807282021266004510100774x^2 + (\text{sym.}) \\
Q_{11} &= 76575720951x^4 + 466015525084217238173676x^3 \\
&\quad - 216521978405797871634733654x^2 + (\text{sym.}) \\
Q_{13} &= 722610532225x^4 + 3464286958371072692766958316x^3 \\
&\quad + 4603575719671472165025576604518x^2 + (\text{sym.})
\end{aligned}$$

each irreducible over $\mathbb{Q}$ with pairwise distinct roots (exact factorization and squarefreeness certificates). In particular $J = 1$ and Theorem 6.21 applies sharply at these levels. (Computational; `first_power_descent.py`. Independently, safe-parameter numerical certification extends the integer polynomials and their irreducibility to $n = 15$ ($h = 8$) and $n = 17$ ($h = 4$), at working precisions up to 2600 digits with at least 168 spare digits per coefficient; `uf_integer_polynomial.py`, `uf_irreducibility.py`.)

Remark 6.23 (Galois groups, as predicted). At every computed level the Galois group of Q_n is the generalized dihedral group $\text{Cl} \rtimes \{\pm 1\}$ of the class group in its regular action — e.g. the regular Klein four-group at $n = 11, 13, 17$ (where Q_n is a V_4 -quartic, hence, like $x^4 + 1$, reducible modulo every prime while irreducible over $\mathbb{Q}$), D_4 at $n = 9$, and an order-16 dihedral image at $n = 15$; the observed Frobenius cycle types match the predicted ones exactly, and $\text{disc } Q_n$ is a square precisely when the predicted image is even. (Computational; `uf_irreducibility.py`.)

Open problem 6.24. Prove $J = 1$ (pairwise distinctness of the u_f) for every odd n . This is the only gap between the per-level irreducibility above and an all- n theorem; compare Theorem 7.16, where the Euclidean analogue is closed by an archimedean dominance argument.

6.7. Denominators are Gross–Zagier primes. The phases are not algebraic integers: already Q_5 has leading coefficient $6647 = 17^2 \cdot 23$. Where the denominators live is governed by the arithmetic of singular moduli. For a discriminant D and $e \in \{-3, -4\}$, write $\text{GZ}(D, e)$ for the set of primes dividing one of the integers $\frac{|e||D|-x^2}{4}$, $x^2 < |e||D|$, $x^2 \equiv |e||D| \pmod{2}$ — by the singular moduli theorem of Gross–Zagier [6], extended to arbitrary discriminant pairs by Lauter–Viray [13], these are exactly the primes at which a CM point of discriminant D can collide, in the sense of Deuring reduction, with the elliptic point of discriminant e (i.e. with $j = 0$ for $e = -3$, $j = 1728$ for $e = -4$).

Proposition 6.25 (the denominator mechanism). *Write $j_i := j(\mathbf{b}_i)$, so $j_1 = \beta_1$ and $j_2 = \beta_2$ is the singular modulus of the class $[\mathbf{r}_n f]$ (Proposition 6.7), and write the sixth power of the phase in the closed form*

$$u_f^6 = \varepsilon^6 \mu^{-12} \frac{j_1^4(j_1 - 1728)^3}{j_2^4(j_2 - 1728)^3} \cdot \frac{\Delta(\mathbf{b}_1)}{\Delta(\mathbf{b}_2)}$$

(from $h_2^6 = C j^4(j - 1728)^3 \Delta$ applied to Proposition 6.10). Then: ε is a unit; the ideal $(\mu) = \mathfrak{m}_2 \mathfrak{m}_1^{-1}$ is supported over the primes dividing the two curvatures (Lemma 6.11); the Δ -quotient is an S -unit supported over the primes dividing $r_0 s_0$ and the curvatures (by the classical index-divisibility of Δ -quotients of nested lattices [12, Ch. 12], applied through $\mathbf{b}_2 = \nu \mathbf{r}^{-1} \mathbf{b}_1$); the numerators $j_1^4(j_1 - 1728)^3$ are algebraic integers; and the prime support of the algebraic integer $j_2^4(j_2 - 1728)^3$ lies in $\text{GZ}(D, -3) \cup \text{GZ}(D, -4)$ [6, 13]. Consequently every prime of the denominator ideal of u_f^6 divides $2 r_0 s_0 q_1 q_2$ or lies in $\text{GZ}(D, -3) \cup \text{GZ}(D, -4)$.

Proposition 6.26 (computational: the support is exactly Gross–Zagier). *For every odd $3 \leq n \leq 17$, every prime of the leading coefficient d'_n of Q_n lies in $\text{GZ}(1 - n^2, -3) \cup \text{GZ}(1 - n^2, -4)$, with the following tagging (an entry $[e]$ means the prime lies in $\text{GZ}(D, e)$):*

n	d'_n	factorization and tags
3	1	—
5	6647	$17^2 [-3] \cdot 23 [-4]$
7	11891	$11 [-3, -4] \cdot 23 [-4] \cdot 47 [-4]$
9	10565574794063311	$11 [-3, -4] \cdot 17^4 [-3] \cdot 19 \cdot 31 [-4] \cdot 59^2 [-3] \cdot 71 \cdot 79 [-4]$
11	76575720951	$3^4 [-3, -4] \cdot 41^2 [-3] \cdot 71 [-4] \cdot 89^2 [-3]$
13	722610532225	$5^2 [-3] \cdot 19^2 \cdot 47 [-4] \cdot 101^2 [-3] \cdot 167 [-4]$
17	819697933195874886721	$23 \cdot 47 [-3, -4] \cdot 71 [-4] \cdot 167^3 [-3, -4] \cdot 191^2 [-3] \cdot 239 \cdot 263 [-4]$

(at $n = 15$ the 33-digit leading coefficient is likewise fully tagged), and in each certified pair-denominator the $\text{GZ}(D, -3)$ -primes enter squared and the $\text{GZ}(D, -4)$ -primes to the first power — the $2 : 1$ ratio reflecting the exponents $(4, 3)$ of $\beta^4(\beta - 1728)^3$ in u^6 ; multiplicities can add across pairs. (Scripts `gz.denominators.py`, `uf.integer_polynomial.py`.)

Thus the denominator of the Schmidt phase is supported exactly at the primes where the level- n CM points collide with the two elliptic fixed points of the modular orbifold — the same two points whose stabilizers produce the $\frac{1}{2}, \frac{1}{3}$ weights in the census $3H(n^2 - 1)$. The exact valuations, the elimination of the residual window of Proposition 6.25, and the fact (a theorem in the sequel) that the Δ -quotient part is always a *unit*, are taken up in Paper II [8].

6.8. The sign of u_f on ambiguous classes. On an ambiguous class u_f is a nonzero real number (Theorem 6.6), and its sign is constant on $\langle \mathfrak{r}_n \rangle$ -cosets by law 2. The sign is *not* a character:

Proposition 6.27 (computational: refutation of the character hypothesis). *No map of the form $\text{sign}(u_f) = -\psi(f)$ with ψ a homomorphism $\text{Cl}[2] \rightarrow \{\pm 1\}$ — in particular no genus character, conductor-aware or otherwise — reproduces the computed signs. At $n = 41$, exact Gauss composition gives $[(3, 0, 140)] \cdot [(4, 0, 105)] = [(12, 0, 35)]$ in $\text{Cl}(-1680)$, while the certified signs give $\psi([3]) = -1$, $\psi([4]) = +1$, $\psi([12]) = +1$: the would-be character fails multiplicativity. ($-\text{sign}(u)$ is multiplicative at all levels ≤ 29 , at 33–39, and at 101 — which is why the hypothesis survives small data — and fails exactly at $n = 31, 41$.) (Signs certified from 140-digit phases at every ambiguous class of every odd $n \leq 41$ and $n = 101$; `make_phase_atlas.py`.)*

What the sign actually is, on the divisor classes, is an archimedean *inequality*. The ambiguous classes with $b = 0$ are the *divisor classes* $f = (a, 0, c)$, $ac = r_0 s_0$, $\gcd(a, c) = 1$: each prime power of $r_0 s_0$ goes wholly to a or to c , so

$$a = a_r a_s, \quad c = c_r c_s, \quad r_0 = a_r c_r, \quad s_0 = a_s c_s, \quad a_r = \gcd(a, r_0), \text{ etc.}$$

Theorem 6.28 (sign law on divisor classes). *For every odd $n \geq 3$ and every ambiguous class $f = (a, 0, c)$ of discriminant $1 - n^2$,*

$$\text{sign}(u_f) = -\text{sign}(a_r c_s - a_s c_r),$$

and $a_r c_s = a_s c_r$ never happens. Equivalently: $u_f > 0$ iff the twisted divisor $a_r c_s$ is $< \sqrt{N}/2$.

Proof. By Theorem 6.18 and Lemma 6.13, $u_f = -r_0 h_2(\mathfrak{b}_1)/h_2(\mathfrak{r}^{-1}\mathfrak{b}_1)$ with $\mathfrak{b}_1 = [1, m_1]$, $m_1 = i\sqrt{N}/(2a)$.

(i) *The twisted lattice, exactly.* In the basis $(1, \omega_0)$, the module $(2a)\mathfrak{r}\mathfrak{b}_1$ is generated by $(2ar_0, 0)$, $(0, 2r_0)$, $(0, 2a)$, $(-2r_0 s_0, 0)$. The ω_0 -column gcd is $2\gcd(r_0, a) = 2a_r$ (contributed by vectors with vanishing first coordinate, so no mixing), and the first-coordinate gcd of the remaining generators is

$\gcd(2ar_0, 2r_0s_0) = 2r_0a_s$. Hence $\mathfrak{r}^{-1}\mathfrak{b}_1 = \frac{1}{r_0}\mathfrak{r}\mathfrak{b}_1 = \lambda[1, \tau_2]$ with $\lambda = a_s/a \in \mathbb{Q}_{>0}$ and

$$\tau_2 = iy_2, \quad y_2 = \frac{2a_r}{2r_0a_s} \cdot \frac{\sqrt{N}}{2} = \frac{a_r\sqrt{r_0s_0}}{r_0a_s} = \sqrt{\frac{a_rc_s}{a_sc_r}}.$$

(ii) *Positivity of the kernel above i.* $F = E_4^2 E_6 / \Delta$ is real on the imaginary axis; $E_4(iy) > 0$ and $\Delta(iy) > 0$ termwise, and $E_6(iy)$ is strictly increasing in y with $E_6(i) = 0$ (by the weight-6 functional equation under $z \mapsto -1/z$), so $F(iy) > 0$ for $y > 1$.

(iii) *Assembling.* $m_1 = iy_1$ with $y_1 = \sqrt{N}/(2a) > 1$ (for a reduced divisor class $a < c$ strictly, since $\gcd(a, c) = 1$ excludes $a = c$), so $h_2(\mathfrak{b}_1) = CF(iy_1)$ with $F(iy_1) > 0$. By homogeneity $h_2(\mathfrak{r}^{-1}\mathfrak{b}_1) = \lambda^{-2}CF(iy_2)$, and by the weight-2 functional equation $F(iy_2) = -y_2^{-2}F(i/y_2)$ when $y_2 < 1$. Thus $\text{sign}(u_f) = -\text{sign} F(iy_2)$, which is -1 if $y_2 > 1$ and $+1$ if $y_2 < 1$, i.e. $-\text{sign}(a_rc_s - a_sc_r)$ by (i). Finally $a_rc_s = a_sc_r$ would force (by pairwise coprimality of the four factors) $a_r = c_r = 1$ and $a_s = c_s = \sqrt{s_0}$ with $r_0 = 1$, i.e. $n = 3$, $s_0 = 2 = a_s^2$ — impossible. $\square$

The sign records on which side of the E_6 -zero locus (the $\text{SL}_2(\mathbb{Z})$ -orbit of i) the $\mathfrak{r}_n$ -twisted CM point falls. An inequality between divisors is not multiplicative — which explains Proposition 6.27 structurally. The theorem recovers the old observations: the principal class has $y_2 = \sqrt{s_0/r_0} > 1$, sign $-$; the $\mathfrak{r}_n$ -class has $y_2 = \sqrt{r_0s_0} > 1$, sign $-$; a $+$ requires genuinely mixed divisors. The predicted sign matches the computed one at every divisor-type ambiguous class of every odd $n \leq 41$ and of $n = 101$ (81 classes; `make_phase_atlas.py`). On the remaining ambiguous classes — types (a, a, c) and (a, b, a) over the ramified prime 2, present only at $n \equiv \pm 1 \pmod{8}$ — a closed form is not yet extracted (Section 8).

6.9. The phase is not cycle-integral data: the Duke–Imamoğlu–Tóth comparison. The level- n data lives on the real quadratic side through the Cartan geodesic: trace $-2n$, length $2 \log \varepsilon_n$, discriminant $d = 4(n^2 - 1) = -4D$. The alignment is exact: the minimal solution of $t^2 - du^2 = 4$ is $(t, u) = (2n, 1)$, so the fundamental automorph of every form of discriminant d is ε_n itself — the unit that normalizes the phase. The invariants classically attached to such d are the cycle integrals of Duke, Imamoğlu and Tóth [3]: for each form class A of discriminant d , $C_A(f) = \int_{\Gamma_Q \backslash S_Q} f(z) \frac{\sqrt{d} dz}{Q(z, 1)}$ for $f = j - 744$, together with the Kronecker-limit integrals E_A (the same integral for $f = \log(\text{Im}(z)^6 |\Delta(z)|)$) and the Rademacher symbols $\Psi(\gamma_A)$ of the automorphs; and the coupled pair $(D, d) = (1 - n^2, 4(n^2 - 1))$ of discriminants of opposite sign is precisely the index shape of the two-sign Fourier coefficients in the Katok–Sarnak framework [10]. It is natural to ask whether the phase is a linear shadow of these invariants. It is not:

Proposition 6.29 (computational: certified non-relation). *For every odd $3 \leq n \leq 17$, compute all cycle integrals $C_A(j - 744)$ and E_A over the*

Phase atlas, level $n = 9$ ($D = -80$, $h = 4$): hue = $\arg u$ (red = $u < 0$, cyan = $u > 0$); +/- = sign on ambiguous classes;
thin segments join τ -twin pairs ($u \rightarrow 1/u$: mirrored hue, inverted shade); dotted curves = mirror geodesics; dotted edge = imprimitive, gray = elliptic-core

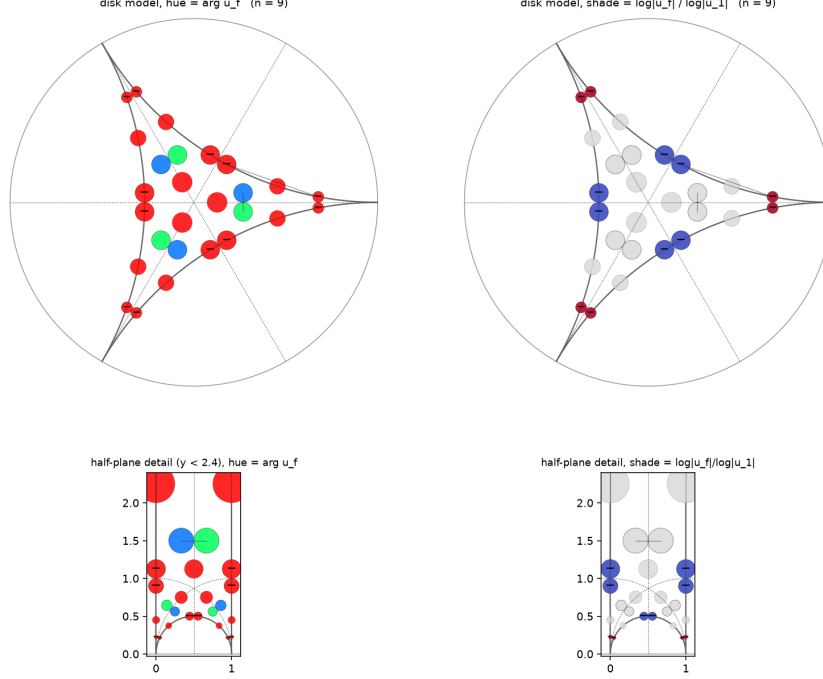


FIGURE 3. The phase portrait of level $n = 9$ (discriminant -80 , class group $\mathbb{Z}/4$): the circles of the ideal triangle colored by $\arg u_f$ (hue) and $\log |u_f| / \log |u_1|$ (shade), with mirror geodesics overlaid. Level 9 is the first level with non-real phases. Mirror-image circles carry conjugate hues (law 1); τ_n -twin pairs carry mirrored hues and inverted shades (law 2). Produced by `make_phase_atlas.py`.

$\mathrm{SL}_2(\mathbb{Z})$ -classes of discriminant $4(n^2 - 1)$ (from the definitions, validated by $C_A(1) = 2 \log \varepsilon_A$, independent parametrizations, and $\mathrm{SL}_2(\mathbb{Z})$ -invariance to 10^{-140}), and the phases u_f at 170 digits. Integer-relation searches (PSLQ) at 150-digit working precision, accepting a candidate only when $(\text{terms}) \times (\text{coefficient digits})$ is far below the working precision, find **no relation** in any of the following tests: $\log |u_f|$ against $[1, \log \varepsilon, \mathrm{Tr}_d(j-744), \sum_A E_A, \pi^2]$ (coefficient bound 10^8); $\log |u_f|$ against $[1, \log \varepsilon]$ (bound 10^{10} ; so u_f is not $\pm \text{rational} \times \varepsilon^k$ in disguise); $\log |u_f|$ against the per-class $\mathrm{Re} C_A(j-744)$ and E_A (bound 10^6); the argument $\arg u_f$ against $[\pi, \log \varepsilon]$, and against the per-class $\mathrm{Im} C_A(j-744)$ and π ; and the pair-sums S_x against the aggregate basis — across the full battery, 52 certified non-relations, zero unsafe candidates, and the only target-involving hit is the proved anchor $u = -1$ at $n = 3$.

Moreover the Rademacher symbols on the trace- $2n$ classes are antisymmetric under class inversion (so $\sum_A \Psi_A = 0$ identically), while $\text{sign}(u_f)$ lives on the imaginary-quadratic 2-torsion and is not even a character: no per-class matching is possible. (Script `dit_comparison.py`; re-run for this paper.)

This is the expected outcome, and the structural reason deserves record: u_f is algebraic (Theorem 6.19), while cycle integrals of j are expected to be transcendental and are not known to be logarithms of algebraic numbers; a nontrivial linear identity would have been a transcendence miracle. The value of the certified mismatch is the license it gives: the phase is genuinely new real-quadratic-adjacent data at $d = 4(n^2 - 1)$, not repackaged cycle-integral material. What the two theories *do* share — exactly ε_n , the coupled pair $(D, -4D)$, and the E_6 -geometry at the elliptic point (Theorem 6.28) — marks the structured place to look next: mixed objects indexed by a pair of discriminants of opposite sign, in the sense of [10], rather than the one-discriminant invariants tested here.

7. THE EUCLIDEAN PHASE

The phase of Section 6 was a two-sided $\text{SL}_2(\mathbb{Z})$ -invariant. Replacing the left factor by the *Euclidean symmetries of the plane inside the Bianchi group* — the translations $N = \{T_\alpha = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} : \alpha \in \mathbb{Z}[i]\}$ — produces a parallel theory with an even cleaner mechanism, and, remarkably, with complete answers: its level polynomials are monic and irreducible for every level. We study functions of X invariant under $X \mapsto T_\alpha X \gamma'$, $\alpha \in \mathbb{Z}[i]$, $\gamma' \in \text{SL}_2(\mathbb{Z})$.

7.1. The moduli problem and the sixth invariant. Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{C})$ with $c \neq 0$ and $X(\mathbb{H})$ a *bounded disk* D ; boundedness is equivalent to $\kappa := 2\text{Im}(\bar{c}d) > 0$, equivalently $w := d/c = -X^{-1}(\infty) \in \mathbb{H}$, and κ is then the curvature of ∂D . For $X \in \text{SL}_2(\mathbb{Z}[i])$ this is the positive-curvature normalization of a Schmidt disk: $\kappa = 2n$, center $\zeta/2n$, $\zeta \equiv i \pmod{2}$. Two invariants are visible at once: the circle modulo $\mathbb{Z}[i]$ -translation (the Cartan datum; three real parameters) and

$$\beta(X) := j(w) = j(-X^{-1}(\infty))$$

(two more; left translations do not touch the bottom row, and right multiplication moves w by an element of $\text{SL}_2(\mathbb{Z})$). Five real parameters in all. What they miss: the double coset of a generic X is described by the oriented lattice $\Lambda = \mathbb{Z}c + \mathbb{Z}d \subset \mathbb{C}$ together with the center of the disk mod $\mathbb{Z}[i]$; of the lattice, the five invariants see the covolume ($= \kappa/2$) and the homothety class ($= j(w)$) but *not the phase of the homothety*. That missing dimension is the sixth invariant:

Definition 7.1. $\Theta(X) := \frac{j'(w)}{c^2} = - \text{Res}_{z=X^{-1}(\infty)} X(z) \cdot j'(-X^{-1}(\infty)).$

Proposition 7.2. $\Theta(T_\alpha X \gamma') = \Theta(X)$ for every $\alpha \in \mathbb{C}$ (not just $\mathbb{Z}[i]$) and every $\gamma' \in \mathrm{SL}_2(\mathbb{Z})$. In lattice terms, $\Theta(X) = \hat{h}_2(\Lambda_X)$, where $\hat{h}_2(\mathbb{Z}c + \mathbb{Z}d) := j'(d/c)/c^2$ is the weight-2 homogeneous extension of j' to oriented lattices: the sixth invariant is the weight-2 value at the actual lattice, not just at its homothety class. Fixing the five invariants, the identity component of the fiber is $X\{h_t\}$ with h_t the rotation group fixing the pole; along it $|\Theta|$ is constant and $\arg \Theta$ moves linearly at rate 2 (the weight). Finally, the full stabilizer of ∞ in $\mathrm{PSL}_2(\mathbb{Z}[i])$ is generated by N and $z \mapsto -z$, which sends $D \mapsto -D$ and $\Theta \mapsto -\Theta$: the two disks $\pm D$ carry opposite phases.

Proof. Left translations do not touch (c, d) . On the right, $(c, d) \mapsto (cp + dr, cq + ds)$, so w moves by an $\mathrm{SL}_2(\mathbb{Z})$ -Möbius map and $c \mapsto c(p + rw)$; the weight-2 law for j' cancels the factor $(p + rw)^2$. Basis-independence of $\hat{h}_2$ is this right-invariance. For the fiber: a real h preserves the circle exactly iff it fixes the pole; the generator of that elliptic group has row eigenvector $(1, w)$ with unit-modulus eigenvalue e^{-it} , so the bottom row $(c, d) = c(1, w)$ transforms by $c \mapsto ce^{-it}$, fixing w and $|c|$ and rotating $\Theta = j'(w)/c^2$ at rate 2. The Borel sign is the computation $\mathrm{diag}(i, -i)$: $(c, d) \mapsto (-ic, -id)$, $c^2 \mapsto -c^2$. $\square$

This is the exact structural analogue of the hyperbolic phase: there, two kernels (one per $\mathrm{SL}_2(\mathbb{Z})$ -factor) glued by a derivative; here, one kernel glued by a residue — the left factor is a translation group, whose derivative is 1 and which needs no kernel.

7.2. Structure of a level: conductor = curvature. Let $K = \mathbb{Q}(i)$, $\mathcal{O}_n = \mathbb{Z} + n\mathbb{Z}[i]$ the order of conductor n and discriminant $-4n^2$, $h(-4n^2)$ its class number.

Lemma 7.3 (disk $\leftrightarrow$ lattice). $D \mapsto \Lambda_D = \mathbb{Z}c + \mathbb{Z}d$ is a bijection from Schmidt disks of curvature $2n$ modulo $\mathbb{Z}[i]$ -translation onto the primitive index- n sublattices $\Lambda \subseteq \mathbb{Z}[i]$ (primitive: $\Lambda\mathbb{Z}[i] = \mathbb{Z}[i]$).

Proof. D determines X up to $X \mapsto T_\lambda X h$, $h \in \pm\mathrm{SL}_2(\mathbb{Z})$, so the oriented lattice is well defined; it has index $|\mathrm{Im}(\bar{c}d)| = n$ and is primitive because $ad - bc = 1 \in (c, d)$. Conversely an oriented basis of a primitive Λ has $\mathrm{gcd}_{\mathbb{Z}[i]}(c, d) = 1$, hence completes to $X \in \mathrm{SL}_2(\mathbb{Z}[i])$, and the completions form exactly one translation class (replacing the top row by $(a + tc, b + td)$ translates the disk by t). $\square$

Lemma 7.4 (conductor = curvature). Every primitive index- n sublattice $\Lambda \subseteq \mathbb{Z}[i]$ is a proper $\mathcal{O}_n$ -lattice: its multiplier ring is exactly $\mathbb{Z} + n\mathbb{Z}[i]$.

Proof. Primitivity forces $\mathbb{Z}[i]/\Lambda$ cyclic, so $\Lambda = \ker \phi$ for a surjection $\phi(x + yi) = sx + ty$ onto $\mathbb{Z}/n$, $\mathrm{gcd}(s, t, n) = 1$. Eliminating the scalar by which $x + yi$ acts on the cyclic quotient gives $x + yi \in \mathcal{O}(\Lambda) \iff n \mid y(s^2 + t^2)$, so the conductor is $n/\mathrm{gcd}(n, s^2 + t^2)$. For $p \mid n$: $p \mid s^2 + t^2$ iff the line $\ker(\phi \bmod p)$ is an ideal line of $\mathbb{Z}[i]/p$, and primitivity excludes exactly the ideal lines; hence $\mathrm{gcd}(n, s^2 + t^2) = 1$. $\square$

(The count of surviving lines — $p + 1$ minus $0/1/2$ ideal lines at inert/ramified/split p — re-derives $N_e(n)$ of Proposition 3.1 from scratch.)

Lemma 7.5 (every class, exactly twice). *For $n \geq 2$, the class map $\Lambda \mapsto [\Lambda] \in \text{Cl}(\mathcal{O}_n)$ from primitive index- n sublattices is surjective with all fibers $\{\Lambda, i\Lambda\}$ of size two; the corresponding disk pairs are $\{D, -D\}$.*

Proof. Given a class, pick an integral representative $\mathfrak{a}$ prime to n and a generator α of $\mathfrak{a}\mathbb{Z}[i]$ ($\mathbb{Z}[i]$ is a PID); then $\Lambda = \alpha^{-1}\mathfrak{a} \subseteq \mathbb{Z}[i]$ is primitive, proper for $\mathcal{O}_n$, of index n (extension of ideals prime to the conductor preserves norms), and the construction is independent of choices up to μ_4 , with $-\Lambda = \Lambda$ and $i\Lambda \neq \Lambda$ (else $i \in \mathcal{O}(\Lambda) = \mathcal{O}_n$). Conversely every primitive Λ arises from its own class. Finally $i\Lambda$ is the lattice of $-D$, and homothety by i does not change the class. $\square$

Theorem 7.6 (Euclidean structure). *For $n \geq 2$, the assignment $D \mapsto [\Lambda_D]$ induces a bijection*

$$\{\text{curvature-}2n \text{ Schmidt disks in } [0, 1) + [0, 1)i\} / (z \mapsto -z) \xrightarrow{\sim} \text{Cl}(\mathbb{Z} + n\mathbb{Z}[i]),$$

and in particular $N_e(n) = 2h(-4n^2)$ for $n \geq 2$ (and $N_e(1) = h(-4) = 1$). Under the bijection, complex conjugation of disks is inversion in the class group, and the disks tangent to $\mathbb{R}$ (the Ford row, $\zeta = \pm i$) are the principal class, with $w \sim ni$.

For $K = \mathbb{Q}(i)$ this is equivalent to Stange's theorem [21] that circles of curvature $f\sqrt{-\Delta}$, up to translation and rotation by 180° , are in bijection with $\ker(\text{Pic}(\mathcal{O}_f) \rightarrow \text{Pic}(\mathcal{O}_K))$ — for class number one, all of $\text{Pic}(\mathcal{O}_f)$. The proof above (Lemmas 7.3–7.5) gives the explicit dictionary the phase theory needs. The count identity reconciles the multiplicative formula of Proposition 3.1 with the class number formula for orders: *the Euclidean census of the Gaussian Schmidt arrangement is a class number census of the non-maximal orders of $\mathbb{Q}(i)$.*

Theorem 7.7 (singular moduli of the Euclidean family). *For $n \geq 2$,*

$$\prod_{\substack{D \text{ Schmidt disk of curvature } 2n \\ \text{center} \in [0, 1) + [0, 1)i}} (x - j(-X_D^{-1}(\infty))) = H_{-4n^2}(x)^2 \in \mathbb{Z}[x],$$

the square of the ring class polynomial of $\mathbb{Z} + n\mathbb{Z}[i]$; for $n = 1$ it is $H_{-4}(x) = x - 1728$.

Proof. $w = d/c$ is a CM point of $\mathbb{Q}(i)$ whose order is exactly $\mathcal{O}_n$ by Lemma 7.4, and $j(w) = j([\Lambda_D])$; by Theorem 7.6 the disks of the level sweep each class exactly twice. $\square$

The anchors are classical: the Ford disks see $j(i) = 1728$; the curvature-4 disks see $j(2i) = 66^3$. At hyperbolic level n the β -values were singular moduli of discriminant $1 - n^2$ (fixed maximal-order type, field varying with the level); at Euclidean curvature $2n$ they are singular moduli of conductor n over the

fixed field $\mathbb{Q}(i)$: the arrangement sees, along its two natural gradings, the two orthogonal aspects of complex multiplication — discriminant and conductor.

Proposition 7.8 (traces: the even-square slice; computational). *Set $\text{Tr}_n^E := \sum_{\mathfrak{c}} (j(\mathfrak{c}) - 744)$ over $\text{Cl}(\mathcal{O}_n)$ for $n \geq 2$, and $\text{Tr}_1^E := 492$; then $\sum_{g|n} \text{Tr}_{n/g}^E = t(4n^2)$, Zagier's trace [23]; verified for all $n \leq 13$ by `euclidean_moduli_invariants.py`. The Euclidean census reads the weight-3/2 coefficients along the even squares $d = 4n^2$, the complementary diagonal to the hyperbolic slice.*

7.3. The lemniscatic phase. On integral X the five invariants produce ring class integers; the sixth is transcendental — but by exactly one period. Define

$$\Omega := \frac{\Gamma(\frac{1}{4})^4}{8\pi^2} = \frac{\varpi^2}{\pi} = ((2\pi)^6 \eta(i)^{24})^{1/6} = 2.18843961\dots, \quad u(D) := \frac{\Theta(X_D)}{\Omega},$$

where $\varpi = 2 \int_0^1 dt / \sqrt{1-t^4}$ is the lemniscate constant — the period of the CM curve $\mathbb{C}/\mathbb{Z}[i]$, whose homothety class is the unique class at $n = 1$. From $j'^6 = -(2\pi)^6 j^4 (j - 1728)^3 \Delta$,

$$(7.1) \quad u(D)^6 = -\beta^4 (\beta - 1728)^3 \frac{\Delta(\Lambda_D)}{\Delta(\mathbb{Z}[i])}, \quad \beta = j(w_D),$$

where $\Delta(\Lambda)$ is the weight-12 lattice discriminant. The Δ -quotient of commensurable CM lattices is algebraic and $n^{12} \Delta(\Lambda) / \Delta(\mathbb{Z}[i])$ is an algebraic integer [12, Ch. 12], so $u(D)$ is algebraic and $n^2 u(D)$ an algebraic integer.

Example 7.9 (the anchor $n = 2$: an exact integer phase). For the curvature-4 disks, $\beta = 66^3$, $\beta - 1728 = 2^3 3^6 7^2$, and $\Delta(\Lambda) / \Delta(\mathbb{Z}[i]) = \eta(2i)^{24} / \eta(i)^{24} = 2^{-9}$, giving $u^2 = -2^4 3^{10} 7^2 11^4$ exactly:

$$u = \pm 2^2 3^5 \cdot 7 \cdot 11^2 i = \pm 823284 i.$$

This is the Euclidean analogue of the hyperbolic anchor $u = -1$ at $n = 3$ — with the characteristic difference that the Euclidean phase is an S -integer rather than a unit.

Proposition 7.10 (laws and the reality criterion). *For every Schmidt disk D of curvature $2n \geq 4$:*

- (1) $u(-D) = -u(D)$: the two disks of a class carry opposite phases, so u^2 is well defined on $\text{Cl}(\mathcal{O}_n)$; write $v_{\mathfrak{c}}^2$.
- (2) $u(\bar{D}) = \overline{u(D)}$ and $[\Lambda_{\bar{D}}] = [\Lambda_D]^{-1}$: conjugation of disks is inversion in the class group.
- (3) On an ambiguous class, $u \in \mathbb{R} \cup i\mathbb{R}$; precisely, with $\zeta = x + yi$ the curvature-center of either disk of the class,

$$u \in \mathbb{R} \iff n \mid y, \quad u \in i\mathbb{R} \iff n \mid x,$$

and for $n \geq 2$ exactly one of the two holds on each ambiguous class. Since y is odd, even levels have no real classes; at every odd $n \geq 3$ the class of $\zeta = (n+1) + ni$ is real.

Proof. (1) is the Borel sign of Proposition 7.2. (2) $\bar{D} = Y(\mathbb{H})$ for $Y = \bar{X} \operatorname{diag}(i, -i)$, with invariant point $w_Y = -\bar{w}$; the real-kernel identity $j'(-\bar{\tau}) = -\overline{j'(\tau)}$ combines the factors to $\Theta(Y) = \overline{\Theta(X)}$, and $\bar{\Lambda}$ represents the conjugate, i.e. inverse, class. (3) Ambiguity means $\bar{\zeta} \equiv \pm\zeta \pmod{2n}$, i.e. $n \mid y$ resp. $n \mid x$; then (2) and (1) give $\bar{u} = u$ resp. $\bar{u} = -u$. Both together would force $n^2 \mid x^2 + y^2 \equiv 1 \pmod{4n}$, impossible for $n \geq 2$. Solvability at $y = n$: $x = n + 1$ works for every odd n . $\square$

7.4. One level, one monic integer polynomial.

Theorem 7.11 (sixth powers, unconditionally). *Fix $n \geq 2$ and let $v_{\mathfrak{c}}$ ($\mathfrak{c} \in \operatorname{Cl}(\mathcal{O}_n)$) be the phases, each taken at either disk of its class. Then*

$$P_n^{(6)}(y) := \prod_{\mathfrak{c} \in \operatorname{Cl}(\mathcal{O}_n)} (y - (n^2 v_{\mathfrak{c}})^6) \in \mathbb{Z}[y],$$

monic of degree $h(-4n^2)$, and the product over all $2h$ disks of the level is $P_n^{(6)}(y)^2$.

Proof. (i) *Rationality.* For $\tau \in \mathbb{H}$ and a cyclic index- n sublattice $\Lambda_{a,b,d} = \mathbb{Z}(a\tau + b) + \mathbb{Z}d \subset \mathbb{Z}\tau + \mathbb{Z}$ ($ad = n$, $\gcd(a, b, d) = 1$), the quantity

$$F_{a,b,d}(\tau) := \frac{n^{12} \hat{h}_2(\Lambda_{a,b,d})^6}{(2\pi)^6 \Delta(\tau)} = -\frac{n^{12}}{d^{12}} j_w^4 (j_w - 1728)^3 \frac{\Delta(w)}{\Delta(\tau)}, \quad w = \frac{a\tau + b}{d},$$

is one member of a Hecke orbit: $\operatorname{SL}_2(\mathbb{Z})$ permutes the $F_{a,b,d}$, so every elementary symmetric function of them (jointly with the j_w) is a meromorphic modular function, holomorphic off the cusp, with rational q -expansion (the root-of-unity coefficients cancel in the symmetrization over b , exactly as for the classical modular polynomials) — hence a polynomial in $j(\tau)$ with rational coefficients. Evaluating at $\tau = i$ ($j = 1728$) makes all symmetric functions of the cyclic family rational, jointly with the β 's. (ii) *The primitive cut.* By Lemma 7.4 the primitive sublattices are exactly the cyclic ones whose β has discriminant $-4n^2$; the discriminant of β 's order is a Galois invariant, so $\operatorname{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$, which permutes the joint pairs $(\beta_\Lambda, F_\Lambda)$, preserves the primitive sub-multiset, whose symmetric functions are therefore rational. Each class appears twice with the same sixth power. (iii) *Integrality.* β and $\beta - 1728$ are algebraic integers and $n^{12} \Delta(\Lambda) / \Delta(\mathbb{Z}[i])$ is an algebraic integer; a monic rational polynomial with algebraic-integer roots lies in $\mathbb{Z}[y]$. $\square$

Contrast with Theorem 6.21: same shape — one level, one integer polynomial — but the Euclidean polynomial is *monic*: the hyperbolic denominators came from kernel values at two CM points in a ratio; here there is one kernel value, its elliptic-point factors $\beta^4(\beta - 1728)^3$ sit in the numerator, and all denominators live over n , cleared by n^2 .

Experimentally the descent to *squares* holds at every computed level with a smaller normalizer: let λ_n be the least positive integer with $P_n^{(2)}(y) :=$

$\prod_{\mathfrak{c}}(y - (\lambda_n v_{\mathfrak{c}})^2) \in \mathbb{Z}[y]$. Certified values (`euclidean_moduli_invariants.py`, 140–840 digits, absolute-error criterion) include

$$P_2^{(2)}(y) = y + 2^4 3^{10} 7^2 11^4 \quad (\lambda_2 = 1),$$

$$P_3^{(2)}(y) = y^2 + 194359980365709312 y - \cdots \quad (\lambda_3 = 1),$$

with

n	2	3	4	5	6	7	8	9	10	11	12	13
$h(-4n^2)$	1	2	2	2	4	4	4	6	4	6	8	6
λ_n	1	1	4	25	1	49	2^5	3^3	25	11^2	4	13^2

The rationality and irreducibility at square level are in fact theorems for every n , by the following three steps.

Lemma 7.12 (the root-free closed form). $u_{\mathfrak{c}}^2 = -12 \beta_{\mathfrak{c}} (\beta_{\mathfrak{c}} - 1728) \frac{g_2(\Lambda_{\mathfrak{c}})}{g_2(\mathbb{Z}[i])}$ — a ring class integer times a weight-balanced ratio of Weierstrass g_2 -values at the pair $\Lambda_{\mathfrak{c}} \subset \mathbb{Z}[i]$, with no root of any modular form extracted anywhere.

Proof. $j'(\tau)^2 = -(2\pi)^2 j(\tau)(j(\tau) - 1728)E_4(\tau)$ from $j' = -2\pi i E_4^2 E_6 / \Delta$ and $E_4^3 = j\Delta$, $E_6^2 = (j - 1728)\Delta$; and $\Omega^2 = (2\pi)^2 \eta(i)^8 = (2\pi)^2 E_4(i)/12$ since $\gamma_2(i) = 12$. Homogenize. $\square$

Lemma 7.13 (transport of balanced pair-values). *Let $F(\Lambda; L)$ be a rational expression in g_2, g_3 of two lattices, isobaric of total weight 0 under joint scaling. For $\sigma \in \text{Aut}(\mathbb{C}/K)$ with idèle s and any pair of K -lattices $\Lambda \subseteq L$ of index n :*

$$\sigma(F(\Lambda; L)) = F(s^{-1}\Lambda; s^{-1}L),$$

the adelic action preserving index, multiplier orders, and (up to the class of s) proper-ideal classes.

Proof. Let $E = E_{\Lambda}$, $E' = E_L$ be the Weierstrass curves and $\phi : E \rightarrow E'$ the isogeny that is analytically $z \mapsto z$, so $\phi^* \omega' = \omega$ for the Weierstrass differentials. By the main theorem of complex multiplication in Shimura's formulation for arbitrary lattices [19, Thm. 5.4], E^{σ} is analytically $\mathbb{C}/s^{-1}\Lambda$ via a parametrization compatible with σ on torsion, and likewise E'^{σ} ; compatibility on the dense torsion forces ϕ^{σ} to be analytically $z \mapsto z$. Writing Λ_1, L_1 for the exact period lattices of the transformed Weierstrass equations, $\Lambda_1 = \theta_E s^{-1}\Lambda$ and $L_1 = \theta_{E'} s^{-1}L$, and ϕ^{σ} is multiplication by $\theta_{E'} \theta_E^{-1}$; but σ commutes with algebraic pullback of the algebraically defined ω , so $(\phi^{\sigma})^* \omega_{L_1} = \omega_{\Lambda_1}$, i.e. $\theta_{E'} = \theta_E$: the transformed pair is *jointly* homothetic to $(s^{-1}\Lambda, s^{-1}L)$. Joint scaling invariance of F finishes. $\square$

Proposition 7.14 (translation law at square level; no cocycle). *For $\sigma \in \text{Gal}(\bar{\mathbb{Q}}/K)$ with Artin class $\mathfrak{c}(\sigma) \in \text{Cl}(\mathcal{O}_n) \cong \text{Gal}(H_n/K)$:*

$$\sigma(u_{\mathfrak{c}}^2) = u_{\mathfrak{c}(\sigma)^{-1}\mathfrak{c}}^2;$$

in particular $u_{\mathfrak{c}}^2 \in H_n$, and with the mirror law (Proposition 7.10) the full group $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ permutes $\{u_{\mathfrak{c}}^2\}$ through the generalized dihedral action.

Proof. Lemma 7.12 exhibits $u_{\mathfrak{c}}^2$ as $\beta(\beta - 1728)$ times $F(\Lambda_{\mathfrak{c}}; \mathbb{Z}[i])$ with $F = -12 g_2(\Lambda)/g_2(L)$, isobaric of joint weight 0. Apply σ : the β -factor translates by ring class reciprocity [2, Thm. 11.36], and Lemma 7.13 gives $\sigma F = F(s^{-1}\Lambda_{\mathfrak{c}}; s^{-1}\mathbb{Z}[i])$. Rescale jointly by a generator ν of the principal ideal $(s^{-1}\mathbb{Z}[i])^{-1}$: $\Lambda' = \nu s^{-1}\Lambda_{\mathfrak{c}}$ is primitive of index n with $[\Lambda'] = \mathfrak{c}(\sigma)^{-1}\mathfrak{c}$, and $\sigma F = F(\Lambda'; \mathbb{Z}[i])$, the u^2 -datum of that class. The representative Λ' is determined up to μ_4 , and F has weight 4 in Λ alone, so $i^{-4} = 1$ kills the ambiguity: at square level there is no cocycle at all. $\square$

Lemma 7.15 (archimedean dominance of the principal class). *For every $n \geq 2$ and every class $\mathfrak{c} \neq 1$: $|u_{\mathfrak{c}}| < |u_1|$; explicitly $|u_{\mathfrak{c}}|/|u_1| \leq 10e^{-\pi n}$.*

Proof. By Proposition 7.2, $|u_{\mathfrak{c}}| = |j'(\tau_{\mathfrak{c}})|/(A_{\mathfrak{c}}\Omega)$ at the reduced CM point $\tau_{\mathfrak{c}} = \frac{-B+2ni}{2A}$, with $A_1 = 1$ and $A_{\mathfrak{c}} \geq 2$ for $\mathfrak{c} \neq 1$ (a reduced form of even discriminant with $A = 1$ has $B = 0$, hence is principal). On the fundamental domain $|q| \leq e^{-\pi\sqrt{3}} < \frac{1}{230}$, and the crude bounds $\sigma_3(k) \leq k^4$, $\sigma_5(k) \leq k^6$ give $|E_4| \leq 2.12$, $|E_6| \leq 3.9$, $|\Delta_q| \geq 0.90|q|$, so $|j'| \leq 2\pi \cdot 19.5 e^{2\pi y}$; at $\tau = ni$ ($q = e^{-2\pi n} > 0$, $n \geq 2$): $E_4 \geq 1$, $E_6 \geq 0.998$, $\Delta_q \leq q$, so $|j'(ni)| \geq 2\pi \cdot 0.998 e^{2\pi n}$. Hence $|u_{\mathfrak{c}}|/|u_1| \leq \frac{19.5}{0.998} \cdot \frac{1}{A} e^{2\pi n(\frac{1}{A}-1)} \leq 10e^{-\pi n} < \frac{1}{50}$. $\square$

Theorem 7.16 (irreducibility at every level). *For every $n \geq 2$:*

- (1) $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ permutes $\{u_{\mathfrak{c}}^2\}$ generalized dihedrally; hence $\prod_{\mathfrak{c}}(y - (n^2 v_{\mathfrak{c}})^2)$ is a monic integer polynomial for all n .
- (2) $P_n^{(2)}$ is irreducible over $\mathbb{Q}$ (in any rational normalization of the roots): the square-phases of a level form a single Galois orbit, $[\mathbb{Q}(v_{\mathfrak{c}}^2) : \mathbb{Q}] = h(-4n^2)$ with $v_{\mathfrak{c}}^2 \in H_n$, and the Galois image is contained in (at the computed levels: equal to) the generalized dihedral group of $\text{Cl}(\mathcal{O}_n)$.
- (3) The same holds for $P_n^{(6)}$.

Proof. (1) Stability under $\text{Gal}(\bar{\mathbb{Q}}/K)$ is Proposition 7.14; complex conjugation is the mirror law; any other σ is a composite. Symmetric functions are then rational, and the roots $(n^2 v_{\mathfrak{c}})^2$ are algebraic integers. (2) By Artin surjectivity the translations act transitively, so all $u_{\mathfrak{c}}^2$ are conjugate and $P_n^{(2)} = m^{|T|}$ with m irreducible and $T = \{\mathfrak{t} : u_{\mathfrak{t}}^2 = u_1^2\}$ a subgroup (the coincidence relation is translation-invariant). Any $\mathfrak{t} \in T$ has $|u_{\mathfrak{t}}| = |u_1|$, so Lemma 7.15 forces $T = 1$. (3) Cube the translation law; the same dominance argument applies. $\square$

The mechanism deserves note. The two hypotheses that remain open on the hyperbolic side (Open problem 6.24 and the historical first-power descent) both evaporate here: the descent because the square-phase has the root-free g_2 -form, and $T = 1$ because the coincidence set is a subgroup, so it only had to be tested against the archimedeanly dominant principal class — and the principal class is the row of disks tangent to $\widehat{\mathbb{R}}$, whose CM point ni escapes to the cusp while every other class stays at height $\leq n/2$. The tangency geometry of the arrangement pins the factorization type.

Remark 7.17 (first powers; computational, open in general). The disk-level multiset $\{\lambda_n u(D)\}$ is $\{\pm \lambda_n v_c\}$, the root set of $P_n^{(2)}(x^2) \in \mathbb{Z}[x]$ of degree $2h = N_e(n)$. At every computed level $2 \leq n \leq 16$ this polynomial is *irreducible* (exact factorization; `mass_law_and_irreducibility.py`): the $N_e(n)$ normalized phases of a level — one per disk in the unit square — form a single Galois orbit and $[\mathbb{Q}(\lambda_n v_c) : \mathbb{Q}] = N_e(n)$; passing from v^2 to v always doubles the degree. Whether this first-power irreducibility holds for all n is open: the natural attack is a weight-2 refinement of Lemma 7.13, where the μ_4 -ambiguity survives as a genuine $\pm$ -cocycle tied to the orientation data — the exact Euclidean shadow of the hyperbolic first-power problem, one power lower than what Theorem 7.16 settles.

7.5. The Δ -mass law. The proof of Theorem 7.16 controls the Galois theory of a level; the following exact product law controls its absolute values. Define the Δ -mass

$$M(n) := \prod_{\mathfrak{c} \in \text{Cl}(\mathcal{O}_n)} n^{12} \frac{\Delta(\Lambda_{\mathfrak{c}})}{\Delta(\mathbb{Z}[i])}.$$

Theorem 7.18 (Δ -mass law). *For every $n \geq 2$,*

$$M(n) = \varepsilon(n) \prod_{\substack{p^k \parallel n \\ p \text{ not split}}} p^{\frac{6}{e_p} \cdot \frac{p^k - 1}{p - 1} N_e(n/p^k)}, \quad \varepsilon(n) = \begin{cases} -1 & n = 2^k, \ k \geq 2, \\ +1 & \text{otherwise,} \end{cases}$$

with $e_2 = 2$ (ramified) and $e_p = 1$ (inert); split primes contribute nothing at all — e.g. $M(5) = M(13) = M(25) = 1$ exactly.

Proof. Four steps; each is separately machine-verified by the script `mass_law_and_irreducibility.py`.

(M1) *The full mass is a constant.* For $\tau \in \mathbb{H}$ let $A(n)(\tau) := \prod_{\Lambda} \Delta(\Lambda) / \Delta(L_{\tau})$, the product over all $\sigma(n)$ index- n sublattices Λ of $L_{\tau} = \mathbb{Z}\tau + \mathbb{Z}$. The family is lattice-intrinsic and Δ -ratios are homothety-invariant, so $A(n)$ is $\text{SL}_2(\mathbb{Z})$ -invariant; in the Hermite parametrization each factor is holomorphic and nonvanishing on $\mathbb{H}$ and meromorphic at the cusp, so $A(n)$ is a rational function of j with no zeros or poles in $\mathbb{C}$ — a constant, equal to its $q \rightarrow 0$ limit. The factor (a, b, d) contributes leading term $d^{-12} \zeta_d^b q^{a/d-1}$; the q -exponents cancel ($\sum_{ad=n} (a - d) = 0$) and $\zeta_d^{0+1+\dots+(d-1)} = (-1)^{d-1}$, so

$$A(n) = (-1)^{t(n)} \prod_{d|n} d^{-12d}, \quad t(n) = \#\{d \mid n : d \text{ even}\}.$$

(M2) *Stratification over $\mathbb{Z}[i]$, and the recursion.* Put $\tau = i$. For $\Lambda \subseteq \mathbb{Z}[i]$ of index n , the ideal $\mathfrak{d} := \Lambda \mathbb{Z}[i]$ is principal, $= (\delta)$, and $\Lambda = \mathfrak{d} \Lambda''$ with Λ'' primitive of index $n/N\mathfrak{d}$; the assignment $\Lambda \mapsto (\mathfrak{d}, \Lambda'')$ is a bijection and $\Delta(\delta \Lambda'') = \delta^{-12} \Delta(\Lambda'')$. Writing $Q(k)$ for the product of $\Delta(\Lambda) / \Delta(\mathbb{Z}[i])$ over primitive index- k sublattices, $r(m)$ for the number of ideals of norm m , and

$\gamma(m)$ for a generator of $\prod_{N\mathfrak{d}=m} \mathfrak{d}$ (so $\gamma(m)^{12}$ is well defined and rational, the ideal product being conjugation-stable), the constant of (M1) factors as

$$A(n) = \prod_{m|n} \gamma(m)^{-12N_e(n/m)} Q(n/m)^{r(m)}.$$

The $m = 1$ stratum carries $Q(n)$ to the first power, so the recursion determines $Q(n) \in \mathbb{Q}$ inductively from $Q(1) = 1$ — the rationality of the mass needs no CM theory — and $Q(n) = [\prod_{\mathfrak{c}} \Delta(\Lambda_{\mathfrak{c}}) / \Delta(\mathbb{Z}[i])]^2 \geq 0$ because $\Delta(i\Lambda) = \Delta(\Lambda)$ pairs the two lattices of each class. Hence $|M(n)| = n^{12h} \sqrt{Q(n)}$ is pinned by the recursion: it suffices to check that the claimed value satisfies it.

(M3) *The exponents.* Both sides are units away from n . Fix $p \mid n$, $n = p^k n'$. Every ingredient factors along p : $v_p(\gamma(p^j)^{12}) = 6j(j+1)$, $6j[2 \mid j]$, $6j$ in the split, inert, ramified cases; $v_p(A(n)) = -12(\sum_{j \leq k} j p^j) \sigma(n')$; and r, N_e are multiplicative with $r * N_e = \sigma$ (Dirichlet convolution — the counting shadow of the stratification, since $r = 1 * \chi_{-4}$ and $N_e = \text{Id} * \mu \chi_{-4}$). Substituting the ansatz $v_p(Q(p^j m)) = v_p(Q(p^j)) N_e(m)$ ($p \nmid m$) — which the claimed closed form satisfies — the recursion factors as a p -power recursion times $r * N_e = \sigma$ over n' , and reduces to $n = p^k$. There, in generating functions, the recursion is a single rational-function identity in each of the three splitting cases, each a one-line check; e.g. split ($\chi = 1$):

$$\frac{-12pT}{(1-T)(1-pT)^2} + \frac{12T}{(1-T)^3} \cdot \frac{1-T}{1-pT} = \frac{1}{(1-T)^2} \cdot \frac{-12(p-1)T}{(1-pT)^2},$$

verifying that the exponent series claimed for the split case satisfies the recursion; the inert and ramified cases reduce to the numerator identities $-p(1-T^2) + T(1-pT) = T-p$ and $-4(1-T) + (1-2T) = 2T-3$. After the n^{12h} -normalization these are exactly the exponents of the theorem, proving that $|M(n)|$ equals the claimed product.

(M4) *The sign.* Conjugation pairs the factor of $\mathfrak{c}$ with that of $\mathfrak{c}^{-1}$, so $\text{sign } M(n)$ is the product over ambiguous classes of the signs of their (real) factors. On a class with $u \in i\mathbb{R}$ every lattice is conjugation-stable, of Hermite shape $\langle d, b + ai \rangle$ with $b \in \{0, d/2\}$, and $\Delta(\Lambda)$ is real with the sign of $e^{2\pi i b/d}$: positive for $b = 0$, negative for $b = d/2$. On a class with $u \in \mathbb{R}$ ($\bar{\Lambda} = i\Lambda$; odd n), the lattice $(1-i)\Lambda$ is conjugation-stable and cannot be rectangular (else $\Lambda \subseteq (1+i)\mathbb{Z}[i]$, contradicting primitivity), so $\Delta((1-i)\Lambda) < 0$ and $\Delta(\Lambda) = -64\Delta((1-i)\Lambda) > 0$. Completing the count: $b = 0$ lattices lie in the $x = 0$ classes ($2^{\omega(n)}$ of them), and $b = d/2$ occurs iff $x \equiv n$, whose class count is 0 unless $4 \mid n$ (a mod-8 obstruction) and $2^{\omega(m)}$ for $n = 2^k m$ ($k \geq 2$, m odd). So $\text{sign } M(n) = (-1)^{2^{\omega(m)}}$ for $4 \mid n$ and $+1$ otherwise: -1 exactly when $m = 1$, i.e. $n = 2^k$, $k \geq 2$. $\square$

Direct evaluation confirms the law through $n = 49$: e.g. $M(2) = 2^3$, $M(4) = -2^9$, $M(6) = 2^{12} 3^{12}$, $M(7) = 7^6$, $M(16) = -2^{45}$, $M(27) = 3^{78}$, $M(49) = 7^{48}$, and $M(5) = M(13) = M(25) = 1$. The proof is elementary,

but the shape of the exponents keeps a conceptual reading: split p is ordinary reduction of the lemniscatic curve (the class-group product cancels along the horizontal isogeny volcano), non-split p is supersingular, where the geometric series $\frac{p^k-1}{p-1}$ is precisely the growth of quasi-canonical-lifting valuations. The per-class refinement — the distribution of the total valuation over the classes, hence the fine structure of the λ_n -table — is taken up in Paper II [8].

7.6. What changed between the two phase theories.

	hyperbolic (§6)	Euclidean (§7)
left/right groups	$\mathrm{SL}_2(\mathbb{Z}) \times \mathrm{SL}_2(\mathbb{Z})$	translations $\mathbb{Z}[i] \times \mathrm{SL}_2(\mathbb{Z})$
CM data of a level	disc $1 - n^2$, varying field	fixed field $\mathbb{Q}(i)$, conductor n
count of a level	$3H(n^2 - 1)$ (weighted)	$N_e(n) = 2h(-4n^2)$ (exact)
j -polynomial	ring class H_{1-n^2}	ring class squared $H_{-4n^2}^2$
trace slice	$t(n^2 - 1)$	$t(4n^2)$
sixth invariant	two kernels + derivative	one kernel + residue
fiber rate	$\sqrt{n^2 - 1}$	2 (the weight)
normalization	$\varepsilon = n + \sqrt{n^2 - 1}$	$\Omega = \varpi^2/\pi$
level polynomial	integer, non-monic (GZ)	monic integer
irreducibility	every computed level	every level (Thm. 7.16)
twist law	$u_f u_{\tau f} = 1$	none needed (Δ -mass)
sign/reality law	divisor classes (Thm. 6.28)	proved center criterion

8. OPEN PROBLEMS

We collect the open ends flagged above, in the order they occurred.

- (1) *Distinctness* (Open problem 6.24): prove the u_f of a level pairwise distinct for every odd n ; with Theorem 6.21 this makes Q_n irreducible for all n . The Euclidean analogue was closed by archimedean dominance (Lemma 7.15); the natural attack is the same estimate at discriminant $1 - n^2$.
- (2) *Exact denominator valuations* (Section 6.7): the support is Gross–Zagier; the exact valuations, and the elimination of the residual window of Proposition 6.25, require the conductor-degenerate refinements of [6, 13] and are the subject of Paper II [8], together with the unit theorem for the Δ -quotient part.
- (3) *The 2-adic sign law* (Section 6.8): extend Theorem 6.28 to the ambiguous classes over the ramified prime ((a, a, c) and (a, b, a) , present at $n \equiv \pm 1 \pmod{8}$); the same Hermite-normal-form mechanism applies but the closed form is not yet extracted.
- (4) *Imprimitive strata* (Remark 4.11): a clean class formula at the ramified prime 2; and the observed phase theory of the imprimitive strata (exact roots of unity at small core class numbers) awaits a framework.
- (5) *Euclidean first powers* (Remark 7.17): irreducibility of $P_n^{(2)}(x^2)$ for all n — a $\pm$ -cocycle refinement of Lemma 7.13.

- (6) *Even levels*: the entire hyperbolic phase theory was run at odd n (the family $\mathcal{S}$); even levels live in $i\mathcal{S}$ with odd discriminants $1 - n^2 \equiv 3 \pmod{4}$ and should transport cleanly.
- (7) *The level $\alpha = 2$* : the excluded level (centers at the elliptic point ρ , $j'(\rho) = 0$) should carry a regularized phase via leading Laurent coefficients.

Beyond these, the arrangement's monoid structure (the half-plane Bianchi matrices form a rigid monoid whose atoms are an Apollonian gasket), its spectral geometry, the spherical phase theory, and the Kronecker-limit evaluation of $\sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log |u|$ against $L'(0, \chi)$ are developed in the repository documents and in Paper II [8]; none of that material is used here.

APPENDIX A. MACHINE VERIFICATION

Every finite claim of this paper — exact identities, class-by-class laws, integer polynomial coefficients, sign tables, and the non-relation battery — is verified by a named script in the repository accompanying the paper. The scripts are standalone Python programs (`python3`, with `mpmath` [9] and `sympy` [15]); each was re-run from a clean checkout for this draft. Runtimes below are on a single commodity core.

Three certification guard rails are enforced throughout. (i) *Exactness first*: wherever a statement is finite, it is checked in exact integer or rational arithmetic (Hermite normal forms, `fractions`, resultants), not floating point. (ii) *Absolute-error criterion*: a numerical value is accepted as an integer or rational only with at least $\max(20, \text{dps}/5)$ spare digits in the absolute-error sense; a huge value with an $O(1)$ fractional part is rejected regardless of relative precision. (iii) *Integer-relation safety*: a PSLQ fit is trusted only when $(\text{number of terms}) \times (\text{coefficient digits})$ is far below the working precision; every reported non-relation logs its margin. Working precisions are set at run time, never at module import.

script	verifies	precision	time
<code>verify_classification.py</code>	Thm. 2.1 (orbit = congruence classes, $n \leq 15$); Prop. 3.1 ($n \leq 200$); Prop. 3.2 (to 10^6)	exact	2.4 s
<code>alpha_circles.py</code>	Thm. 3.3 ($2 \leq n \leq 40$; spot 301, 1000)	exact	0.1 s
<code>spherical_moduli_invariants.py</code>	Thm. 3.6 ($\ell \leq 20$); Prop. 3.5	exact, 40d	0.3 s
<code>involution_experiments.py</code>	(4.1); fixed points; Prop. 4.4	exact	1.0 s
<code>involution_classmap.py</code>	class map, all classes, odd $n \leq 41$	exact	0.1 s
<code>proof_check.py</code>	Lemmas 4.6–4.8 + Thm. 4.5, 169 classes	exact	0.1 s
<code>composition_check.py</code>	§5 recipes, 1913 pairs, odd $n \leq 41$	exact	0.1 s
<code>matrix_composition_check.py</code>	Props. 5.5–5.6; Rem. 5.7	exact	0.1 s
<code>moduli_invariants.py</code>	Lemmas 6.4–6.5, Thm. 6.6, kernel independence ($n \leq 13$)	60–80d	0.1 s
<code>laws</code>	laws 1–2 at every class, odd $n \leq 17$	80d	0.8 s
<code>deep</code>			
<code>first_power_descent.py</code>	exact Φ_m ($m \leq 10$); $\omega_f = 1$ exact (odd $n \leq 21$); Thm. 6.18 numerics (≥ 108 d agreement, $n \leq 13$); Thm. 6.22 (exact Π_n , table match, irreducibility)	exact, 110d	3.8 s
<code>uf_integer_polynomial.py</code>	certified Q_n, Ψ_n (odd $n \leq 17$); GZ tagging	120–2600d	128 s
<code>uf_irreducibility.py</code>	irreducibility of Q_n, Ψ_n , two routes; Galois groups	exact	2.5 s
<code>gz_denominators.py</code>	Prop. 6.26 tagging	exact	0.1 s
<code>euclidean_moduli_invariants.py</code>	§7 structure ($n \leq 24$), H^2 , traces, phases, $P^{(2)}, P^{(6)}$, λ_n , reality patterns, Δ -mass ($n \leq 49$)	exact, 140–840d	4.5 s
<code>mass_law_and_irreducibility.py</code>	Thm. 7.18 (M1–M4; recursion $n \leq 60$); Thm. 7.16 record ($2 \leq n \leq 16$ incl. $P^{(2)}(x^2)$); Lemma 7.15 margins	exact	6.5 s
<code>make_phase_atlas.py</code>	two-route phases (≥ 138 d), laws, Q_n roots; Thm. 6.28 (81 divisor classes, $n \leq 41, 101$); Prop. 6.27	100–140d	0.9 s + 0.9 s
<code>dit_comparison.py</code>	Prop. 6.29: DIT-side validation and the PSLQ battery	150–170d	29 s; battery 30 min

The modular polynomials Φ_m used in Theorems 6.18–6.22 are constructed exactly from integer q -expansions inside `first_power_descent.py` (power sums of $j((a\tau + b)/d)$ collapse to integer series; peeling against powers of j is an identity because a holomorphic $\mathrm{SL}_2(\mathbb{Z})$ -invariant function vanishing at ∞ vanishes, and the zero tail is asserted with slack), validated by integrality, symmetry, the classical Φ_2 , and numerical root checks at 120 digits. The phases enter the atlas and battery through two independent routes — the canonical-matrix closed form of Proposition 6.10 and the twisted-lattice form of Theorem 6.18 — which agree to at least 138 digits at every class of every

atlas level; at $n \leq 17$ every computed phase is additionally a root of the published integer polynomial Q_n to 10^{-140} .

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